Alienorum Calculation Reference

Physical models, empirical estimates, and procedural appearance

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Alienorum starts with catalog measurements and fills gaps with physical models, empirical estimates, and explicit assumptions. This reference follows those calculations from stellar properties and orbits through atmospheres, climates, terrain, and vegetation. Each mathematical expression is placed alongside the question it addresses, the meaning of its symbols, and the limits of its interpretation.

Where a calculation accepts a measured or supplied quantity, that quantity takes priority over an estimate. The distinction matters when looking at a generated world: an illustrative atmosphere or landscape does not establish that an unresolved object actually has oceans, life, particular minerals, or the weather shown on screen.

The equations describe the scientific operations carried out by the application. To reproduce a particular result exactly, consult the corresponding implementation and its inputs. The final appendix gives source locations and file digests that identify the repository state used to build this document. The original scientific audit and this later explanatory revision are dated separately above.

Unless a section states otherwise, quantities use SI units. M denotes mass in kilograms, R radius in metres, g gravitational acceleration in metres per second squared, p pressure in pascals, and T temperature in kelvin. Orbital notation uses a for semimajor axis and e for eccentricity. U denotes an independent random number drawn uniformly from [0,1). Symbols can have another local meaning; those uses are introduced where they occur.

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1. Reading the models and preserving evidence

Source for these calculations: planet.cpp; shore.cpp; numeric.h; savemodels.h.

It helps to distinguish three kinds of statement in the pages that follow. Some relations express conservation, geometry, or equilibrium thermodynamics under stated approximations. Others estimate a property by interpolating observations or published model grids. A third group supplies a workable scenario where the available evidence does not determine a unique answer. Procedural appearance belongs mainly to that third group.

A supplied number remains usable when it carries an uncertainty annotation. Zero is also a valid value and is treated differently from an absent measurement. Pressure, composition, radius, temperature, or water coverage entered in a catalog or editor can therefore constrain an inference or replace it altogether.

Saved or locked appearances may retain parameters from an earlier calculation. As a result, restoring an object can legitimately produce a different state from inferring it afresh from the same catalog entry.

Several compact functions are used throughout the reference. C(x;l,h) restricts x to the interval from l to h. The smoothstep S uses the restricted coordinate u to join two states with a flat slope at each end:

$$C(x;l,h)=\min(h,\max(l,x)),\qquad S(x)=u^2(3-2u),\quad u=C(x;0,1)$$

A logistic transition provides another gradual change between zero and one. The separate notation U(l,h) means a uniform random draw between the lower bound l and upper bound h:

$$\ell(x)=\tfrac{1}{1+e^{-x}},\qquad U(l,h)=l+(h-l)U$$

Clipping places a value between two bounds. Smoothstep and logistic functions make a transition gradual rather than abrupt, while U(l,h) converts a unit-interval draw into a draw between l and h. These operations recur in both physical approximations and appearance rules.

A bound used for rendering may simply keep a numerical calculation stable or a shape visually reasonable. Similarly, a range of scenario settings describes alternative assumptions rather than a statistical confidence interval. Reproducing a realization also calls for its input provenance, diagnostic records, random state, and simulation epoch.

Order of inference

The calculations form a chain. Stellar properties and the orbit determine the incident radiation. Mass and radius then determine gravity and escape speed. The atmosphere follows from assumptions about the original volatile inventory, release from the interior, escape, weathering, and condensation; radiation and phase partition are solved together where the model couples them.

The retained water inventory is distributed over terrain to find a waterline. Local temperature and thermal history determine which parts are liquid or frozen. Cloud and

vegetation scenarios add an appearance, and illumination and viewing geometry determine how that appearance reaches the observer.

For global irradiation, the application follows one resolved stellar light source. A moon uses its parent planet's orbit around that source. Radiation from other stellar companions is not generally added to this calculation. The largest uncertainties can instead come from poorly constrained age, elemental abundances, delivery history, interior storage, vertical mixing, and aerosols.

2. Units, gravity, density, and basic astrometry

Source for these calculations: celestial.cpp; point.cpp; star.cpp; planet.cpp.

Mass and radius are the starting point for several later calculations. Treating the body as a sphere gives its mean density ρ, surface gravitational acceleration g, and escape speed v_{esc}. G is the gravitational constant:

$$\rho = \frac{3M}{4\pi R^3}, \quad g = \frac{GM}{R^2}, \quad v_{esc} = \sqrt{\frac{2GM}{R}}$$

These expressions use kilograms. Many application records, however, store mass in grams and use a correspondingly adjusted gravitational constant. Atmospheric column calculations explicitly convert to kilograms; overlooking that conversion would introduce a factor of one thousand.

The common surface-gravity estimator reports gravity relative to Earth. Calculations of scale height and atmospheric inventories instead use dimensional acceleration.

A flattened body has another geometric distinction: its equatorial radius R_{eq} exceeds its polar radius R_{pol}. Starting from the volume-equivalent radius R and oblateness f, the application uses:

$$R_{eq} = \frac{R}{(1-f)^{1/3}}, \qquad R_{pol} = (1-f)R_{eq}$$

Oblateness f measures how much shorter the polar radius is than the equatorial radius. The expressions distribute the supplied mean radius between those two axes while preserving the volume of an oblate spheroid. They prescribe the shape without solving for the rotating interior that would produce it.

Astrometry locates the object before any local appearance is calculated. With distance d, right ascension α, and declination δ, its Cartesian position vector is:

$$\mathbf{r} = d(\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta)$$

Parallax gives distance through the reciprocal relation below, when parallax ϖ is in arcseconds and distance d_{pc} is in parsecs. Proper motion μ, in arcseconds per year, then gives the transverse speed v_t; the factor 4.74047 converts these astronomical units to kilometres per second:

$$d_{pc} = \frac{1}{\varpi_{arcsec}}, \qquad v_t = 4.74047\,\mu_{arcsec/yr} d_{pc}\ \mathrm{km\,s^{-1}}$$

Right ascension α and declination δ specify the direction to the object; distance supplies the length of the position vector. Coordinate rotations then express that vector in the selected reference plane and view. When proper motion and radial velocity are available, they also constrain space velocity.

Generated populations receive the velocity distributions described later. In either case, linear motion is a kinematic approximation: it does not follow a star through an integrated Galactic gravitational potential.

3. Stellar temperature, luminosity, and spectral approximations

Source for these calculations: star.cpp; stellarevolution.h; wdcolor.h; wdlimb.h.

A stellar color index measures the difference between brightness in two bands. In the blackbody approximation, B−V can be related to the ratio of Planck radiances B_λ at the representative V and B wavelengths. The correction c_{BV} sets the application's color zero point:

$$(B-V)(T) = 2.5\log_{10}[B_{\lambda_V}(T)/B_{\lambda_B}(T)] - c_{BV}$$

To estimate temperature from B−V, the application searches for the temperature whose Planck spectrum gives the corresponding B/V ratio after applying the color zero point c_{BV}. It uses 50 bisection steps over 1000–200000 K and holds the result at an endpoint for colors beyond that range. A supplied temperature or a spectral table can take priority.

Color is an imperfect thermometer. Interstellar reddening, chemical abundance, surface gravity, and an unresolved companion can all change B−V without producing the temperature change implied by a simple blackbody.

To relate a visual magnitude to total luminosity, the bolometric correction BC_V(T) is added to the absolute visual magnitude M_V. The resulting M_{bol} gives luminosity relative to the Sun, using the adopted solar bolometric magnitude of 4.74:

$$M_{bol} = M_V + BC_V(T), \qquad \frac{L}{L_\odot} = 10^{0.4(4.74 - M_{bol})}$$

Radius and effective temperature provide a second route to luminosity through the Stefan–Boltzmann relation. Solar subscripts denote the reference values used to express the ratio:

$$\frac{L}{L_\odot} = \left(\frac{R}{R_\odot}\right)^2 \left(\frac{T}{T_\odot}\right)^4$$

The bolometric correction BC_V translates a visual-band absolute magnitude into a magnitude representing the star's total radiated power. Across the stellar range, the application uses the Flower/Torres polynomial in $\log_{10}(T)$ [R5]. Below 3500 K it continues linearly between −4.33 mag at 2500 K and −1.75 mag at 3500 K. Above 50000 K it extends the boundary value using the ratio of V-band Planck flux to total flux.

Those continuations are approximate, particularly for molecular atmospheres and stellar remnants.

The Planck function supplies the wavelength-dependent radiance used in these color and flux estimates. Here λ is wavelength, h is Planck's constant, c is the speed of light, and k_B is Boltzmann's constant:

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}(e^{hc/(\lambda k_B T)} - 1)^{-1}$$

Main-sequence properties are obtained by interpolation along a tabulated spectral sequence. This supplies typical masses, radii, luminosities, temperatures, and colors, rather than a unique solution for every individual star.

A separate evolution-grid estimate compares normalized logarithmic temperature, luminosity, and metallicity coordinates. It combines 64 nearby models using weighted summaries. Its reported age range describes that procedure; it does not include every uncertainty or possible stellar history.

For white dwarfs, the relation between mass and radius differs from that of ordinary stars. The application uses the following compact approximation, with m equal to mass in solar units and R_{WD} the estimated radius:

$$R_{WD} = 0.0126 R_\odot m^{-1/3}\sqrt{1-(m/1.44)^{4/3}}, \quad m = M/M_\odot$$

In this approximation the mass is capped at 0.99 times 1.44 solar masses, and an absent mass is replaced by 0.6 solar masses. Published white-dwarf atmosphere colors can replace the blackbody color estimate. The atmosphere type and cooling history remain important to the interpretation of either result.

4. Orbital geometry, rotation, and tidal constraints

Source for these calculations: celestial.cpp; planet.cpp; moon.cpp; orbit_spacing.h.

An orbit first requires a clock relating elapsed time to position. For semimajor axis a and combined central and orbiting masses M_c+M, n is mean orbital angular frequency and P is period. Mean anomaly □ advances uniformly; eccentric anomaly E converts that uniform clock to motion along an ellipse:

$$n = \sqrt{\frac{G(M_c + M)}{a^3}}, \quad P = \frac{2\pi}{n}, \quad E - e\sin E = \mathcal{M}$$

Once E is known, the distance r from the focus and the coordinates in the orbital plane follow directly. The x-axis points toward periapsis, while e controls the ellipse's departure from a circle:

$$r = a(1 - e\cos E), \quad (x, y) = a(\cos E - e, \sqrt{1 - e^2}\sin E)$$

The eccentric anomaly E is found numerically from Kepler's equation. The resulting coordinates lie in the orbital plane. Rotations through the argument of periapsis, inclination, and longitude of the ascending node place that ellipse in three-dimensional space.

This follows a Keplerian trajectory. The spacing and clearance filters described below reject some obviously unsuitable arrangements, but they do not demonstrate long-term stability in a system of interacting bodies.

Satellite placement is constrained at both large and small distances. The Hill radius R_H estimates the scale on which the parent can retain satellites against the central body. The fluid Roche radius estimates where tides can disrupt a satellite, using parent radius R_c and parent and satellite densities ρ_c and ρ_s:

$$R_H = a(1 - e)\left(\frac{M}{3M_c}\right)^{1/3}, \quad R_{Roche} \simeq 2.44 R_c (\rho_c/\rho_s)^{1/3}$$

A generated satellite must clear the parent and the applicable Roche limit, while remaining within an allowed fraction of the parent's gravitational sphere of influence. Generated moons extend to at most 0.3 of that outer scale. Ring placement uses a fluid-disruption scenario.

Material strength, irregular shape, orbital resonances, and long-term tidal migration can change real systems. Those effects are not integrated by these placement rules.

Where a rotation period is absent, the generator uses simple tidal-state rules. P_{lock} is an eccentricity-dependent orbital-period threshold. The second expression gives the spin period for the separate 3:2-resonance branch:

$$P_{lock} = \frac{100}{1 + 4e} \text{ days}, \qquad P_{spin} = \frac{2}{3} P_{orb} \quad (3:2)$$

When rotation is unknown, a short orbital period or the hot-giant classification can lead to synchronous rotation: the same face remains directed toward the parent. The threshold P_{lock} is a heuristic, rather than a calculation of the time required for tidal locking.

If the synchronous branch does not apply, an orbital period below 250 days together with e above 0.15 selects a Mercury-like 3:2 resonance. Three rotations then occur during two orbits, so P_{spin} is 2/3 of P_{orb}. The original audit corrected that period ratio.

Other default rotation periods follow A/ln(M), with M inserted numerically in grams and the result expressed in seconds. A is 2.06×10^6 for ocean-rich classes, 1.74×10^6 for ice giants, 1.11×10^6 for gas giants, and 2.38×10^6 for the remaining classes. The gram reference is part of the empirical convention; changing the mass unit without changing the calibration would change the result.

The cycle of daylight depends on rotation relative to the orbit, not relative to distant stars. Subtracting orbital from spin angular frequency gives the solar angular frequency:

$$\omega_{solar} = 2\pi\left(\frac{1}{P_{spin}} - \frac{1}{P_{orb}}\right)$$

The solar angular frequency describes how rapidly the direction of the star moves through the local sky. It depends on the difference between the spin and orbital frequencies, rather than on spin alone.

When the application is minimized, it pauses position and appearance calculations. On restoration, monotonic elapsed time advances the simulation clock under the usual observer time-dilation rule, and orbital positions are evaluated at that new epoch. The idle interval is not applied as a camera-motion step.

5. Incident starlight, albedo, and equilibrium temperature

Source for these calculations: planet.cpp; patch.cpp; star.cpp.

The stellar luminosity and orbital distance set the energy available to heat a world. F_* is the flux at semimajor axis a, F(t) is the instantaneous value at eccentric anomaly E, and the final expression gives the time average over an orbit:

$$F_* = \frac{L/L_\odot}{(a/\mathrm{AU})^2}, \quad F(t) = \frac{F_*}{(1-e\cos E)^2}, \quad \langle F\rangle = \frac{F_*}{\sqrt{1-e^2}}$$

F_* is the irradiation at the stellar semimajor axis, expressed relative to Earth. Multiplying a relative flux by 1361 converts it to W m^{-2}. F(t) includes the changing distance along an eccentric orbit, and the angle brackets denote an average over an orbit.

The atmospheric-retention model uses the orbital average. The baseline radiating temperature uses the semimajor-axis flux, with seasonal distance changes applied afterward. For a moon, that baseline comes from the planet’s orbit around the star.

A reference temperature follows by balancing absorbed starlight against thermal emission from a sphere. A_B is Bond albedo, the fraction of total incident power reflected away, and σ is the Stefan–Boltzmann constant:

$$T_{eq} = \left[\frac{1361 F_*(1-A_B)}{4\sigma}\right]^{1/4}$$

The factor of four accounts for the difference between the disk area intercepting sunlight and the whole spherical area emitting thermal radiation. The expression assumes unit emissivity and global redistribution of absorbed energy. T_{eq} is therefore a reference radiating temperature, rather than the local temperature below the star or a measured surface temperature.

Geometric albedo A_g compares a body's full-phase brightness with that of a perfectly reflecting diffuse disk of the same cross-section. Bond albedo A_B is the fraction of total incident energy reflected across all directions and wavelengths. To estimate A_B from sparse calibration data, the application transforms both albedos and fits their relation with color B−V:

$$y = -\ln(1 - A_g), \quad z = -\ln(1 - A_B), \quad z = b_0 + b_1(B - V) + b_2 y$$

The application estimates Bond albedo by fitting a local plane through the three closest calibration points in the transformed coordinates. It then converts back with $A_B = 1 - \exp(-z)$, clipped to [0,1]. If those points form a degenerate triangle, it uses the nearest point instead.

The eight calibration triples (B−V, A_g, A_B) are (0.97,.142,.088), (.81,.689,.760), (.20,.434,.294), (1.43,.170,.250), (.87,.538,.503), (1.09,.499,.342), (.56,.488,.300), and (.41,.442,.290). This is a sparse empirical interpolation. In particular, extrapolating it does not have the reliability of integrating a measured reflectance spectrum.

An internally heated world can radiate more than it absorbs from the star. If q_{tide} is tidal power per unit surface area, the corresponding radiating temperature T_{rad} is obtained by adding the internal flux to the equilibrium energy balance:

$$T_{rad}^4 = T_{eq}^4 + \frac{q_{tide}}{\sigma}$$

Tidal heat adds an internal energy flux q_{tide} to the absorbed stellar power. Adding the fourth powers before taking the fourth root respects that energy balance. This path does not provide a general model of radiogenic heating or residual formation heat for every object.

6. Tidal heat as an internal energy source

Source for these calculations: tidalheat.h; planet.cpp.

Tidal deformation can turn orbital and rotational energy into heat. The model here uses equilibrium tides, assumes zero spin obliquity, and retains the full eccentricity functions. It estimates heating at the present state without evolving the orbit and interior together.

To shorten the expressions, z denotes e^2, b denotes $1-z$, and x is the spin angular frequency ω divided by orbital mean motion n. The quantities A and B below are eccentricity polynomials. The assumed dissipation ratio k_2/Q is converted through a lag convention at a representative forcing frequency.

The eccentricity dependence is collected into two polynomials, A and B. Writing them explicitly allows the following expressions to retain the full adopted eccentricity dependence rather than just a small-eccentricity approximation:

$$A = 1 + 3z + \frac{3}{8}z^2, \quad B = 1 + \frac{15}{2}z + \frac{45}{8}z^2 + \frac{5}{16}z^3$$

These polynomials set the minimum-dissipation spin ratio x_{eq} and the coefficient Ω that penalizes departure from that ratio:

$$x_{eq} = \frac{B}{Ab^{3/2}}, \quad \Omega = \frac{A}{b^{9/2}}$$

Even at $x=x_{eq}$, an eccentric orbit can dissipate energy. H_{min} is the dimensionless minimum-heating term in the adopted equilibrium-tide formulation:

$$H_{min} = \frac{z[3.5 + 11.25z + 28z^2 + (685/64)z^3 + (255/128)z^4 + (25/512)z^5]}{Ab^{15/2}}$$

The surface heat flux combines that minimum term with the extra dissipation from non-equilibrium spin. M_c is the tide-raising mass, R the heated body's radius, a the orbital semimajor axis, and k_2/Q the assumed tidal response divided by its dissipation quality factor:

$$q_{tide} = \frac{3(k_2/Q)GM_c^2R^3n}{4\pi a^6}\frac{H_{min} + \Omega(x - x_{eq})^2}{\max(1, 2|x-1|)}$$

When spin is absent, the calculation uses x_{eq}, the equilibrium spin giving minimum dissipation in this model. The default k_2/Q is 0.003 for worlds with solid-surface maps and 3×10^{-6} for gaseous worlds. A supplied zero turns this heat source off.

The calculation omits a changing material response, ocean dissipation, obliquity tides, resonant forcing, and variation of Q over time. At high eccentricity it can predict heating too large for a plausible steady interior [R6].

7. Elemental abundances and formation composition

Source for these calculations: chemistry.cpp; salinity.h.

Chemical abundance ratios are conventionally logarithmic. [X/H] compares the number ratio of element X to hydrogen with the same solar ratio. A measurement relative to iron can be converted by adding [Fe/H]:

$$[X/H] = \log_{10}(N_X/N_H) - \log_{10}(N_X/N_H)_\odot, \quad [X/H] = [X/Fe] + [Fe/H]$$

The square-bracket notation compares an elemental abundance ratio with its solar value on a base-ten logarithmic scale. Each abundance bundle retains the solar reference used by its source, and conversions account for offsets between references.

When an element is missing, a clearly marked formation scenario may scale a solar abundance with iron. That supplies an assumed inventory; it does not turn a stellar abundance into a measured planetary bulk composition.

To construct a simple condensate inventory, elemental amounts are normalized to silicon. The logarithmic abundances ε_X and ε_{Si} give n_X, the number of atoms of X per silicon atom. O_{rock} is the oxygen first allocated to the assumed rock-forming mixture:

$$n_X = 10^{\varepsilon_X - \varepsilon_{Si}}, \quad O_{rock} = \min(n_O, n_{Mg} + 2)$$

The remaining oxygen is divided between the assumed carbon-bearing molecules and water. O_C is the allocation to carbon chemistry, and O_W is the nonnegative remainder available to the water proxy:

$$O_C = \min[\max(0, n_O - O_{rock}), 0.65 n_C], \quad O_W = \max(0, n_O - O_{rock} - O_C)$$

The rock proxy combines the iron, magnesium, silicon, and allocated oxygen masses. Numerical coefficients are the corresponding atomic masses; silicon contributes one normalized atom:

$$m_{rock} = 55.845 n_{Fe} + 24.305 n_{Mg} + 28.085 + 15.999 O_{rock}$$

The oxygen remainder gives a water mass proxy m_W, while the carbon abundance gives m_C. Their common normalization allows the water share f_{ice} to be compared with the rock share:

$$m_W = 18.01528 O_W, \quad m_C = 12.011 n_C, \quad f_{ice} = \frac{m_W}{m_W + m_{rock}}$$

The calculation allocates oxygen to silicate rock first. For the carbon-bearing molecules it assumes 45% CO, 10% CO_2, and 45% CH_4, which together consume 0.65 oxygen atoms per carbon atom. The remaining oxygen supplies the water proxy.

Water-to-rock and carbon-to-rock ratios are compared with the corresponding solar calculation to scale the inventory scenarios. Detailed condensation equilibrium, oxidation states, carbon grains, disc transport, and migration are outside this simplified bookkeeping [R7].

A related ratio estimates iron's share of the assumed rocky condensate. The denominator includes the iron, magnesium, silicon, and silicate oxygen assigned by this simplified mixture:

$$f_{Fe} = \frac{55.845 n_{Fe}}{55.845 n_{Fe} + 24.305 n_{Mg} + 28.085 + 15.999(n_{Mg} + 2)}$$

The iron-to-rock quantity is evaluated only within its supported abundance regime. It estimates the iron share of this assumed condensate mixture; it is not an observation of the core mass fraction.

For independent abundance errors, first-order uncertainty propagation uses the variance $\Sigma(\partial f/\partial x_i)^2 s_i^2$, where s_i is the error in input x_i. Correlated errors and systematic offsets between abundance measurements are not included.

8. Delivery inventories and atmospheric retention

Source for these calculations: shore.cpp; shore.h.

The inventory model begins with reference accessible masses equal to 1.45×10^{-4} of planetary mass for CO_2-equivalent material, 10^{-6} for nitrogen, and 2.3×10^{-4} for water. These are terrestrial calibration anchors. They describe material available to the modeled release and storage processes, rather than universal primordial abundances.

The default age is 4.5 Gyr. An inferred stellar age can replace it. Delivery factors then adjust the reference pools according to mass and the assumed formation environment.

The delivery model uses a nominal snow-line scale a_s that grows with stellar mass. The smooth weight w increases for present orbital distance a beyond that scale:

$$a_s = 2.7\, C(M_*/M_\odot; 0.08, 10)\ \text{AU}, \quad w = \frac{1}{1 + (a_s/a)^4}$$

D_W scales water delivery relative to the Earth-distance reference, while D_M gives a weak mass dependence. Planetary and stellar masses are expressed relative to Earth and the Sun in these factors:

$$D_W = \frac{0.001 + 0.02w}{0.001 + 0.02/[1 + 2.7^4]}, \quad D_M = C(M/M_\oplus; 10^{-4}, 100)^{1/4}$$

Carbon and nitrogen use the combined heavy-volatile delivery factor D_H. The separate s_W sets the fraction of water assigned to the surface-accessible reservoir as a function of planetary mass:

$$D_H = D_M \frac{1 + \sqrt{D_W}}{2}, \qquad s_W = \frac{1}{1 + 2\sqrt{M/M_\oplus}}$$

The chemical water and carbon factors multiply their respective reference inventories. Water also receives the mass and distance delivery factors D_M D_W, a factor of three, and the surface-accessible fraction s_W, normalized to one-third at Earth mass. Carbon and nitrogen receive the heavy-volatile factor D_H.

The resulting bulk mass fractions are capped at 0.5 for water, 0.05 for CO_2, and 0.01 for nitrogen. Present orbital distance is only a weak indication of where a world formed, so these factors describe a delivery scenario rather than a reconstructed formation history.

Atmospheric retention can also be summarized by a shoreline score. It compares escape speed, orbit-averaged irradiation, and stellar luminosity with fixed reference scales. The angle brackets on F denote the time average over the orbit:

$$\mathcal{S} = 4\log_{10}\frac{v_{esc}}{5000} - \log_{10}\frac{\langle F\rangle}{0.43} + 0.6\log_{10}\frac{L}{L_\odot}$$

The shoreline score is motivated by the empirical relationship between atmospheric retention and the fourth power of escape speed [R1]. A positive score favors retention under this proxy. It does not specify a unique pressure, and the inventory solver does not interpret it as a universal probability that a planet has an atmosphere.

9. Release, stellar activity, and escape history

Source for these calculations: shore.cpp.

High-energy, or XUV, irradiation here means X-ray and extreme-ultraviolet radiation. Stellar activity in this range is assumed to remain saturated for an initial time t_s. Stellar mass and the metallicity factor A_Z set that duration, bounded to the stated interval:

$$t_s = C[0.1(M_*/M_\odot)^{-2.5}A_Z; 0.01, 5]\ \mathrm{Gyr}$$

The accumulated activity I(t) is the time integral of a constant early level followed by a declining power law. Both age t and saturation duration t_s are expressed in Gyr:

$$I(t) = t\ (t \le t_s), \qquad I(t) = t_s\left[1 + \frac{1-(t/t_s)^{-0.2}}{0.2}\right]\ (t > t_s)$$

The assumed activity history is initially saturated and then decreases as $t^{-1.2}$. The metallicity factor A_Z is 10^(0.3[Fe/H]) only for main-sequence stars of 0.7–1.3 solar masses with a supported metallicity definition. In other cases it is one.

This choice allows a sensitivity to composition within the scenario. It should not be read as a general law of stellar magnetic activity.

D compares integrated activity with a reference 4.5 Gyr history having saturation time 0.1 Gyr. X scales the activity exposure, and the mean factor Ā enters the effective heavy-gas loss coefficient k:

$$D = \frac{I(t;t_s)}{I(4.5;0.1)}X, \quad \bar{A} = \frac{4.5D}{t}, \quad k = \min[10^{12}, 0.035\langle F\rangle\bar{A}(11200/v_{esc})^6]$$

Material release from the interior has its own characteristic time τ_r. The mass scaling below places that time between 0.01 and 2 Gyr:

$$\tau_r = C[0.15\sqrt{M/M_\oplus}; 0.01, 2]\ \mathrm{Gyr}$$

With exponential release and a constant effective loss coefficient k, the fraction retained in the accessible reservoir is f_r. A limiting expression is used when release and loss rates coincide:

$$f_r = \frac{e^{-kt} - e^{-t/\tau_r}}{1 - k\tau_r}, \qquad f_r \to (t/\tau_r) e^{-t/\tau_r} \quad (k\tau_r \to 1)$$

The release model transfers material exponentially from the interior into an atmospheric reservoir while an effective loss process removes it. The retained fraction f_r is the resulting balance of those two rates. Its limiting expression avoids numerical cancellation when the rates approach equality.

The initial inventory is divided into unreleased, retained, and escaped mass. The heavy-gas loss coefficient is calibrated against broad Venus, Earth, and Mars pressure scales; it describes an effective historical loss process rather than energy-limited escape of hydrogen.

Water receives an additional irradiation-dependent loss factor. The logistic function □ creates a narrow, smooth increase around 1.15 times Earth's orbit-averaged flux:

$$k_W = k[1 + 500\ell(((F) - 1.15)/0.025)]$$

For strongly irradiated worlds, the smooth steam transition increases the water-loss rate above the heavy-gas rate. This represents a possible loss history. In the cold-world branch, this water-loss term is zero. At age zero, none of the interior pool has yet been released. The portion assigned to deep storage is removed before the surface condensation calculation.

10. Cold volatile worlds and nitrogen frost

Source for these calculations: shore.cpp; shore.h.

Cold, low-density worlds use a separate treatment because much of their volatile inventory can be ice and nitrogen can condense at the surface. The branch is selected below a bulk density of 3000 kg m^{-3} and below 0.08 times Earth's orbit-averaged irradiation. These thresholds choose a model regime; they do not demonstrate an icy interior.

The cold-world ice fraction is estimated by mixing the specific volumes, or reciprocal densities, of effective rock and ice components. The clipping bounds prevent the density proxy from implying an arbitrarily small or large ice share:

$$f_{ice} = C\left[\frac{1/\rho - 1/3500}{1/1300 - 1/3500}; 0.05, 0.7\right]$$

The density estimate mixes specific volumes using effective rock and ice densities of 3500 and 1300 kg m^{-3}. The resulting f_{ice} becomes the water inventory. Accessible nitrogen is $\min(0.01, 1.35\times10^{-4}\ f_{ice}/0.51$ times the nitrogen abundance scale), while 0.01 of retained water is allocated to the surface.

These choices anchor a Titan-scale scenario. They do not follow a detailed differentiation or outgassing history.

A temperature-like proxy T_x represents the thermal scale for historical nitrogen escape. The dimensionless binding parameter λ compares escape energy for nitrogen, with molar mass 0.028 kg mol⁻¹, against thermal energy. R_g is the molar gas constant:

$$T_x = \max(2.7, 10^4\sqrt{\langle F\rangle \bar{A}}), \quad \lambda = \frac{0.028 v_{esc}^2}{2R_g T_x}$$

The effective nitrogen loss coefficient k_N combines the activity factor Ā with the strong suppression of thermal escape at large λ:

$$k_N = 1000\bar{A}\sqrt{T_x/1000}\,(1+\lambda)e^{-\lambda}\ \mathrm{Gyr}^{-1}$$

T_x represents a historical escape-temperature proxy, and λ compares molecular binding energy with thermal energy. The loss expression has the shape of a Jeans-escape dependence, with its exponent bounded to prevent numerical overflow. The interior release time is 0.1 Gyr.

Neither T_x nor the resulting loss rate is a measurement of the exobase. Magnetospheric sputtering is also absent from this treatment.

At sufficiently low temperature, nitrogen remaining in the accessible inventory can condense. The liquid-branch saturation relation gives p_{N^2} in pascals through a base-ten logarithm referenced to 10^5 Pa:

$$\log_{10}(p_{N_2}/10^5) = 3.7362 - \frac{264.651}{T - 6.788} \quad (T \geq 63.14\ \mathrm{K})$$

Below 63.14 K, the calculation switches to solid-phase vapor-pressure fits. Above 35.61 K the fit is ln(p/bar)=8.514−458.4/T−19870/T²+480000/T³−4524000/T⁴. Below 35.61 K it is 12.4−807.4/T−3926/T²+62970/T³−463300/T⁴+1325000/T⁵. Multiplicative factors join the branches continuously.

Below 10 K an exponential continuation approaches vacuum monotonically. At and above 126.19 K, the nitrogen critical temperature, the saturation cap is removed [R8].

A separate energy-balance approximation estimates the temperature of nitrogen frost. A_f is frost albedo, ε_f its emissivity, and F_* the relative stellar flux. The alpha and beta labels refer to the assumed solid-phase branches:

$$T_\beta = 255\left[\frac{F_*(1-A_f)}{0.7\varepsilon_f}\right]^{1/4}, \quad T_\alpha = 3^{1/4}T_\beta$$

The frost scenario uses reflectivity A_f=0.8 and emissivity ε_f=0.67. It selects the beta-phase temperature above 35.61 K, the alpha-phase temperature below it, and a coexistence plateau at 35.61 K where the two branches would straddle the transition. A supplied cold temperature can replace this estimate.

Nitrogen collision-induced absorption contributes a bounded opacity proportional to p_{N^2}/g. It is handled separately from the terrestrial gas-opacity terms.

11. Surface pressure and coupled gas–condensate partition

Source for these calculations: shore.cpp; planet.cpp.

An accessible gas mass becomes surface pressure through its weight. P_{col} is the pressure scale for unit planetary mass fraction spread over the surface; f_i is the available gas mass fraction of species i, and q_i is its column weight:

$$P_{col} = \frac{GM^2}{4\pi R^4}, \quad q_i = P_{col} f_i, \quad p = \sum_i q_i$$

A mass fraction f_i gives the amount of species i relative to the planet. Multiplying it by P_{col} gives its hydrostatic column weight q_i in pascals, and summing those column weights gives pressure.

This assumes nearly constant gravity across an atmosphere that is thin compared with the planetary radius. It is not a detailed equation of state for a deep envelope. The inferred pressure is also limited to 10^8 Pa as a simulation bound.

The atmosphere cannot retain an unlimited vapor abundance at a given temperature. The saturation pressure is approximated with a constant molar latent heat L_m, reference pressure p_0, reference temperature T_0, and molar gas constant R_g:

$$p_{sat}(T) = p_0 \exp\left[\frac{L_m}{R_g}\left(\frac{1}{T_0} - \frac{1}{T}\right)\right]$$

For water, the reference point is $(T_0, p_0) = (273.16\ \text{K}, 611.657\ \text{Pa})$. The adopted molar latent heat is 51060 J mol^{-1} below that temperature and 45050 above it; the critical temperature is 647.10 K. For CO_2, the corresponding reference is (216.59 K,518500 Pa), with latent heats 25000/16000 J mol^{-1} and critical temperature 304.13 K.

Holding latent heat constant makes these saturation expressions approximate, especially near a critical point. The water saturation pressure is further multiplied by 0.15 to represent effective humidity in a mixed atmospheric column.

For each species, the gas column is the smaller of its available inventory and its saturation-limited column. The conversion from partial pressure to column weight involves molar mass μ_i and the mixture's mean molar mass $\bar{\mu}$, so the equations are solved together:

$$q_i^{gas} = \min\left(q_i^{available}, p_{sat,i}\frac{\mu_i}{\bar{\mu}}\right), \quad \bar{\mu} = \frac{\sum_i q_i^{gas}}{\sum_i q_i^{gas}/\mu_i}$$

Column weight and mole partial pressure are different quantities. Condensation is therefore solved together with the mean molar mass, so that removing one gas does not leave the partial pressures of the others inconsistent.

The inferred reservoirs track CO_2, N_2, and H_2O with molar masses approximately .044, .028, and .018 kg mol^{-1}. Once the gas column weights are known, mole fractions are proportional to q_i/μ_i.

Weathering is strongest in an assumed warm, wet temperature window. The dimensionless weight w_T combines smooth lower and upper temperature transitions with available water mass fraction f_W:

$$w_T = \ell((T - 273.15)/5)[1 - \ell((T - 360)/10)]^{\frac{f_W}{f_W + 10^{-6}}}$$

The fraction of accessible carbon remaining in air then decreases with age t, weathering efficiency η_w, and a bounded mass dependence:

$$f_{C,air} = \exp[-3t\, w_T \eta_w \min(1, \sqrt{M/M_\oplus})]$$

Weathering transfers carbon out of the accessible atmospheric reservoir into a protected pool when the water and temperature factors favor it. The weathering strength is set by η_w, age t, and the mass-dependent factor in the expression.

Because temperature determines partition and partition changes radiative warming, the solver searches for mutually consistent states. It scans 128 intervals for sign changes and refines bracketed solutions by bisection. It favors stable, converged roots near a heuristic climate anchor and retains alternative equilibria and residual errors in the diagnostics. A supplied temperature acts as a constraint even when it disagrees with radiative balance; that disagreement remains visible in the record.

12. Thermal opacity and the Titan calibration

Source for these calculations: misc.cpp; shore.cpp; planet.cpp.

The thermal model summarizes infrared trapping with a gray optical depth τ: a single effective measure replaces the wavelength-dependent absorption spectrum. Let P be pressure in Earth atmospheres, x_i the mole fraction of gas i, and $p_i = P\, x_i$ its partial-pressure proxy in those units.

Each gas contributes a logarithmic term with empirical coefficients a_i and b_i. Hydrogen modifies the methane contribution through h_{H^2}, and additional terms represent absorption associated with molecular collisions. This provides a calibrated warming approximation without a line-by-line radiative-transfer calculation.

Each gas contribution τ_i grows logarithmically with its partial-pressure proxy. The table supplies a_i and b_i; h_{H^2} increases the methane term when hydrogen is present:

$$\tau_i = a_i \ln(1 + b_i p_i), \qquad h_{H_2} = 1 + 25 p_{H_2}$$

Species	a_i	b_i	Sign / qualification
CO_2	.5	100	positive
CH_4	1.2	500 $h(H_2)$	positive
H_2O	.8	50	positive
N_2O	.7	350	positive

Species	a_i	b_i	Sign / qualification
O_3	.4	150	positive
SO_2	.5	80	positive
H_2S	.15	20	positive
CO	.05	10	positive
HCN	.1	30	positive
NH_3	1.7	800	positive
C_2H_6	.20	600	subtracted calibration term

The combined gray optical depth sums those terms, retains the fitted subtraction for ethane, and adds CO_2 and hydrogen collision-induced absorption. The outer maximum keeps the total nonnegative:

$$\tau = \max\left(0, \sum_{i \neq C_2H_6} \tau_i - \tau_{C_2H_6} + 0.00719 p_{CO_2}^2 + \tau_{H_2,\,CIA}\right)$$

Hydrogen collision-induced absorption is applied only above half an Earth atmosphere. The final expression relates total gray optical depth to surface temperature T_s, starting from the stellar-plus-internal radiating temperature T_{rad}:

$$\tau_{H_2,\,CIA} = 0.08 p_{H_2} P \quad (P > 0.5), \qquad T_s = T_{rad}(1 + 1.8\tau)^{1/4}$$

The ethane contribution is subtracted to retain the application’s Titan temperature calibration. Ethane does absorb infrared radiation [R4]. Here the negative sign represents an effective net cooling adjustment within the combined empirical model, rather than a negative molecular absorption coefficient.

The model does not calculate Titan’s photochemical aerosol production or its shortwave anti-greenhouse balance explicitly. Replacing this adjustment with separate gas and aerosol energy budgets would require a new calibration against Titan. The original audit retained the term.

Hydrogen collision-induced absorption turns on at P=0.5, introducing a small discontinuity when H_2 is present. The quadratic CO_2 collision term is calibrated near Venus. These coefficients also incorporate approximately terrestrial gravity, which limits their transfer to very different gravities.

For very large warming $\Delta = T_s - T_{rad}$, the portion above 500 K is compressed through $500 + 100 \tanh((\Delta - 500)/100)$. This keeps extreme extrapolations bounded. It does not identify a physical runaway-greenhouse threshold.

13. Primordial hydrogen–helium envelopes and object classification

Source for these calculations: shore.cpp; chemistry.cpp; planet.cpp; classification_model.json.

A low-density hydrogen–helium envelope can substantially enlarge a planet. The model starts with a rocky-core radius R_c and an assumed initial envelope mass fraction f_0, using dimensionless mass m=M/M⊕:

$$R_c = 1.02 R_\oplus m^{0.27}, \quad f_0 = C[0.015(m/5)^{1.5}; 10^{-5}, 0.2]$$

The envelope thickness R_{env} then depends on retained mass fraction f, irradiation F in Earth units, age t in Gyr, and planetary mass m. The weak irradiation exponent and stronger envelope-fraction exponent come from the fitted relation:

$$R_{env} = 2.06 R_\oplus m^{-0.21} (f/0.05)^{0.59} F^{0.044} (\max(0.1, t)/5)^{-0.18}$$

The dimensionless mass m is M/M⊕. The envelope-thickness relation follows Lopez and Fortney [R2], whose published models cover 1–20 Earth masses, 0.01–20% envelope mass, 0.1–1000 times Earth's irradiation, and ages of 0.1–10 Gyr. Values outside those ranges are approximate extensions of the fitted relation.

To estimate erosion by high-energy radiation, the model assigns an expanded early absorption radius R_X. The integrated XUV energy per unit area, E_{XUV}, produces an energy-limited lost mass fraction Δf at assumed efficiency 0.1:

$$R_X = \max(2R_c, R_c + R_{env}(f_0, 0.1)), \quad \Delta f = \frac{0.1 \pi R_X^3 \mathcal{E}_{XUV}}{GM^2}$$

The XUV exposure is estimated from stellar irradiation F, saturated high-energy fraction 10^{-3}, integrated activity I(t), and activity multiplier X. The time integral is converted from Gyr to seconds to give consistent energy units:

$$\mathcal{E}_{XUV} = 1361 F\, 10^{-3} I(t) X \text{ (Gyr converted to seconds)}$$

A binding parameter λ_H adds an approximate thermal-retention threshold, using an assumed mean molar mass of 0.0023 kg mol⁻¹. The retained envelope fraction f combines the logistic retention factor with the estimated lost fraction:

$$\lambda_H = \frac{GM(0.0023)}{R_g T_{eq} R_X}, \quad f = \max[0, f_0 \ell((\lambda_H - 20)/3) - \Delta f]$$

The early expanded radius, heating efficiency of 0.1, and saturated XUV fraction of 10^{-3} are assumptions of the scenario. The calculation does not couple cooling to mass loss or include core-powered loss and tidal corrections to escape.

The classification score multiplies the initial envelope score by f/(f+.001). A threshold of 0.5 selects a class only where the observational evidence and type locks allow it; a measured radius can override that choice.

For the retained envelope, its weight supplies an approximate pressure at the core surface. Adding the fitted envelope thickness to the core radius gives the total radius:

$$P_{base} = \frac{GM^2 f}{4\pi R_c^4}, \quad R = R_c + R_{env}$$

A separate empirical classifier compares large- and small-radius planets using logistic scores based on $\log_{10}$ mass, $\log_{10}$ irradiation, and metallicity. Its abundance-aware coefficients are (−1.025614,10.724353,−2.900638,2.057835), and the baseline coefficients are (−1.652304,10.917212,−2.717543).

The training labels distinguish radii at least 2 Earth radii from radii at most 1.4 Earth radii. They do not measure atmospheric composition. The calibrated domain spans 2.55–29.32 Earth masses, 2.49–2290 times Earth's irradiation, host temperatures of 4900–6300 K, and metallicities of −.44 to .439 dex. Balancing the training classes means that the scores cannot be interpreted directly as occurrence probabilities. A separate cool-formation assumption limits extrapolation from the hot-planet sample.

14. Bombardment, erosion, and accessible volatile delivery

Source for these calculations: shore.cpp; celestial.cpp.

Impacts can both deliver volatiles and remove material already present. The enhancement B increases as the assumed debris-belt inner edge a_{belt} moves inward. X_{imp} scales the impact exposure, and f_{imp} is cumulative arriving mass as a fraction of the planet:

$$B = C\left[\sqrt{30\text{AU}/a_{belt}}; 1, 4\right], \quad f_{imp} = \min[0.01, 10^{-5}(B-1)X_{imp}t/4.5]$$

If the inner edge of a debris belt is unknown, B is one. The added bombardment term therefore describes an excess above the reference impact history. The assumed arriving material has water, CO_2, and nitrogen mass fractions of .1, .01, and .001 respectively.

A characteristic impact-speed parameter q^2 compares orbital and escape-speed scales. The model uses it to set a capture fraction c and an erosion efficiency ε:

$$q^2 = \min\left(10^6, \frac{GM_*}{av_{esc}^2}\right), \quad c = \frac{1}{1+q^2}, \quad \varepsilon = \min(1, 0.03q^2)$$

Erosion exposure is measured relative to the available gas and mobile-condensate fractions, f_{gas} and f_{mobile}. The survival function V averages the fate of material delivered throughout an interval of loss:

$$d_g = \frac{0.9 f_{imp}\varepsilon}{f_{gas}}, \quad d_s = \frac{0.1 f_{imp}\varepsilon}{f_{mobile}}, \quad V(d) = \frac{1-e^{-d}}{d}$$

The reference partition assigns 90% of the erosion capacity to atmospheric gas and 10% to exposed, mobile condensates. The exposure d sets the fraction of an original reservoir removed through $1-\exp(-d)$. Protected carbon and deep water do not participate in that erosion.

The capture fraction c determines how much arriving material is initially retained. Material delivered throughout the history then survives the combined erosion and escape exposure according to V(d), with continuous limit V(0)=1. Diagnostic records separate prompt loss, later loss, retained delivery, and erosion of the original inventory. These are effective exposure assumptions rather than hydrodynamic impact simulations.

The visible crater population is treated separately. Increasing pressure reduces crater abundance through 1/(1+3p/p⊕), while larger B increases it. Suppressing craters in the appearance does not remove impacts from the physical volatile budget.

15. Water inventory, basin volume, and surface coverage

Source for these calculations: shore.cpp; watercoverage.h; planet.cpp.

The first step from water inventory to surface coverage is to spread the available mass over the entire sphere. The resulting equivalent depth d_W is the depth of a hypothetical uniform water layer, using surface-accessible mass fraction $f_{W,surface}$ and water density ρ_W:

$$d_W = \frac{f_{W,surface} M}{4\pi R^2 \rho_W}, \quad \rho_W = 1000\ \mathrm{kg\,m^{-3}}$$

That uniform depth must be compared with the capacity of the terrain's basins. The model assigns a root-mean-square (RMS) physical relief scale σ_h that decreases with gravity and cannot exceed a fraction of radius:

$$\sigma_h = \min(0.05R, 2500 \times 9.80665/g)$$

The reference physical relief is 2500 m at Earth gravity. It scales inversely with gravity, as a simple material-support scenario, and is capped at 5% of planetary radius. This physical relief is distinct from the vertical exaggeration used to draw terrain: otherwise a shallow decorative landscape could flood with an unrealistically small inventory.

The retained condensed water includes ice. Its area coverage therefore does not establish how much exposed liquid is present.

For a sampled terrain, each location below a trial waterline contributes its depth to basin volume. The first equation matches that volume to d_W; the second counts the equal-area samples lying below the same waterline:

$$d_W = \frac{\sigma_h}{s_h}\frac{1}{N}\sum_j \max(0, h_0 - h_j), \quad f_{area} = \frac{1}{N}\sum_j \mathbf{1}_{h_j < h_0}$$

The samples h_j are normalized terrain heights taken at equal-area locations, and s_h is their standard deviation. The solver adjusts the waterline h_0 by bisection until the summed basin volume matches the available water. The indicator in the area expression counts locations below that level.

A flat surface with any positive water inventory is fully covered. The visual waterline is stored within [0,1], but the represented physical water depth can extend above the highest sampled terrain.

Without a terrain map, a Gaussian elevation distribution supplies a preliminary basin model. In the following relation, z is the standardized waterline, Φ(z) the fraction of area below it, and φ(z) the normal probability density:

$$\frac{d_W}{\sigma_h} = z\Phi(z) + \phi(z), \quad f_{area} = \Phi(z)$$

Before a terrain map exists, the model assumes a Gaussian distribution of elevations, or Gaussian hypsometry. Here Φ is the standard normal cumulative distribution and φ its probability density; z expresses the waterline in units of the relief scale. This supplies an initial coverage and a visual level clipped from 0.5+0.08z.

Actual elevation samples later calibrate the level. The percentage editor measures equal-area coverage, so its percentage is generally different from the numerical waterline setting.

16. Salinity, freezing, boiling, and exposed liquid

Source for these calculations: salinity.h; surfaceclimate.h; surfacephase.h.

Dissolved salt changes the freezing point, so the ocean model estimates salinity before testing phase. The salt-to-water mass ratio r_s compares an Earth-based salt inventory, scaled by Z_s, with the relative available liquid-water supply W:

$$r_s = \frac{35\ Z_s}{965\ W}, \quad S_{ocean} = \min\left(264, \frac{1000 r_s}{1 + r_s}\right)\ \mathrm{g\,kg^{-1}}$$

The salt-inventory multiplier Z_s uses the mean sodium and chlorine abundance relative to iron. Subtracting [Fe/H] removes a uniform solar-scaled metallicity change, and the exponent is bounded:

$$Z_s = 10^{C[0.5([Na/H] + [Cl/H]) - [Fe/H];\ -4,\ 4]}$$

The salinity estimate starts with an Earth-calibrated amount of salt available for mobilization. Z_s adjusts that salt inventory, and W measures the relative liquid-water supply with a positive numerical floor. If sodium is unavailable, magnesium can act as a proxy; chlorine can use sulfur or argon. With no constraining abundance ratios, Z_s is one.

A supplied salinity takes priority. Uniformly scaling a solar mixture with metallicity does not, by itself, make the modeled ocean arbitrarily saltier. The estimate is a bulk salt scenario rather than a chemical analysis of individual dissolved ions.

The freezing temperature T_f depends on both salinity S and pressure. The polynomial gives an approximate seawater relation, with the pressure increment p_{dbar} measured in decibars:

$$T_f = 273.15 - 0.0575S + 0.001710523S^{3/2} - 0.0002154996S^2 - 0.000753p_{dbar}$$

In the seawater polynomial, S is salinity in g kg⁻¹, clipped to 120, and p_{dbar} is the pressure increment above the reference expressed in decibars. At greater salinity, a linear extension approaches the 252.05 K NaCl-brine scenario by 233 g kg⁻¹.

The extension does not replace a full phase diagram. In particular, high-pressure ice forms and the behavior of specific dissolved mixtures are absent.

Boiling is estimated by inverting a saturation-pressure relation. Pressure p is in pascals, and the result T_b is limited to the adopted critical-temperature scale:

$$T_b = \min\left[647.096, \left(\frac{1}{373.15} - \frac{8.314}{40660}\ln\frac{\max(1,p)}{10^5}\right)^{-1}\right]$$

If the inverse-temperature expression becomes nonpositive, the estimate is set to 647.096 K. The constant-latent-heat approximation supplies a pressure-dependent boiling limit, but does not classify a supercritical fluid as liquid water.

The surface model removes exposed ocean liquid where pressure and temperature favor evaporation. An ice-covered reservoir can still contain liquid below the surface. Land frost and exposed water follow separate thermal-response rules.

17. Latitude, altitude, day–night contrast, and regional climate

Source for these calculations: climate.h; planet.cpp.

A global mean temperature does not specify conditions at a particular latitude or altitude. The local model starts with pressure P in Earth atmospheres, contrast scale D, and redistribution factor η. Higher pressure reduces the imposed contrast and increases redistribution:

$$P = p/101325, \quad D = T_m[0.04 + 0.25/(1+P)], \quad \eta = P/(P+0.1)$$

The lapse rate Γ controls cooling with altitude. Separately, T_{lat} redistributes the mean temperature T_m with z, the sine of latitude:

$$\Gamma = 0.0065C(g/g_\oplus; 0, 3)\frac{P}{P+0.01}, \quad T_{lat} = T_m + D(1/3 - z^2)$$

The coordinate z is sine latitude. Its area-weighted mean square over a sphere is 1/3, so $D(1/3-z^2)$ redistributes temperature with latitude without shifting the baseline mean. Later land and seasonal terms can modify that simple balance.

The dimensionless η represents horizontal redistribution, while Γ is a lapse rate in K m⁻¹. Both are approximations. Γ is neither a complete moist adiabat nor a measured lapse rate for an individual atmosphere.

For a world locked to its star, heating follows the illuminated hemisphere. The unit vectors n and s point outward from the surface and toward the star, so their dot product gives the solar projection. f_{int} is the internally supplied fraction of the energy budget:

$$T_{sync} = T_m[f_{int} + (1 - f_{int})(\eta + 4(1 - \eta)\max(0, \mathbf{n} \cdot \mathbf{s}))]^{1/4}$$

On rotating land, a separate term warms an assumed subtropical dry belt. Its latitude dependence is localized rather than applied uniformly across the globe:

$$\Delta T_{dry} = 0.04 T_m \frac{P}{P + 0.01} \left| \max\left(0, 1 - \left(\frac{z^2 - 0.18}{0.2}\right)^2\right) \right|^2$$

Dry-belt warming is added only to non-synchronous land. Regional variation then breaks up otherwise smooth boundaries with a spherical sum of sinusoids. The three weights are .65, .25, and .10, with amplitude $T_m \times .012/(1+P/10)$. For coordinates (x,y,z), the factors are sin(3x+1.7y)sin(4z−2y), sin(13x−7y+3z)sin(11z+5y), and sin(43x+29y)sin(37z−17y).

These fields describe an assumed climate pattern. They are not observed weather. Contrast, dry-belt warming, and regional amplitude are all reduced by $1 - f_{int}$ as internal heat becomes more important. Elevation h above sea level subtracts Γh, and temperature is bounded below by 2.7 K.

18. Seasonal and diurnal thermal response

Source for these calculations: thermalclimate.h; planet.cpp; climate.h.

Temperature responds gradually because the atmosphere, land, and ocean store heat. C_A is atmospheric heat capacity per unit area, using specific heat c_p and column mass p/g. C_L adds land storage and an assumed contribution from nearby ocean coverage f_o:

$$C_A = c_p \frac{p}{g}, \quad C_L = 2 \times 10^6 + C_A + 0.3 f_o \eta (2.09 \times 10^8)$$

Ocean heat capacity C_O includes a larger water reservoir. Dividing the relevant capacity C by radiative response coefficient B gives the thermal relaxation time τ:

$$C_O = C_A + 2.09 \times 10^8, \quad \tau = C/B, \quad B = \max[0.01, (1361 F_*(1 - A_B) + 4 q_{tide})/T_m]$$

Each C is a heat capacity per unit area, in J m^{-2} K^{-1}. B measures the change in outgoing energy flux per kelvin, in W m^{-2} K^{-1}, so C/B is a response time τ in seconds. The ocean term represents roughly 50 m of water, and f_o is the equal-area water coverage.

Atmospheric heat capacity is assembled from the mixture’s mole fractions and molar heat capacities. An absent mixture uses an approximate terrestrial-air value. Mixed-layer depth and heat transport remain scenario assumptions.

The amount of sunlight averaged over a day depends on latitude φ, stellar declination δ, and the sunset hour angle H. Q_d is the average projection onto the illuminated surface:

$$Q_d(\varphi, \delta) = \frac{H \sin\varphi \sin\delta + \cos\varphi \cos\delta \sin H}{\pi}$$

The sunset angle is determined geometrically. Restricting the arccosine argument to [−1,1] also represents the limits of polar night and continuous daylight:

$$H = \arccos C(-\tan\varphi \tan\delta;\ -1, 1)$$

The sunset angle H is clipped to handle continuous polar day and polar night. The resulting daily average is sampled at 64 orbital phases to estimate annual heating.

The temperature history uses 128 samples, each weighted by its duration and by exp(−lag/τ), then normalized to unit total weight. Short thermal memories sample uniformly in time over up to ten response times. Longer memories sample eccentric anomaly with the corresponding duration weights. This periodic quadrature does not depend on the display frame rate.

A periodic forcing can be decomposed into harmonics. For harmonic n with input amplitude $A_{n,in}$, thermal storage reduces the output amplitude and delays its phase by φ_n. The product nω_solarτ measures how slow the thermal response is relative to that harmonic:

$$A_n^{out} = \frac{A_n^{in}}{\sqrt{1 + (n\omega_{solar}\tau)^2}}, \quad \phi_n = \tan^{-1}(n\omega_{solar}\tau)$$

The daytime illumination curve is represented by eight harmonics. For land, the diurnal response time is (2×10⁵+0.05C_A)/B. For water, the heat capacity blends between 4.18×10⁶ and 1.93×10⁶ J m⁻² K⁻¹ over a 263.15–283.15 K liquid transition, with 0.05C_A added.

Changes in stellar illumination modulate the external contribution to T⁴; the internal contribution stays constant. Where mean temperature is supplied, a √(1−e²) normalization prevents orbital eccentricity from silently changing the interpretation of that mean.

19. Sea ice as an enthalpy reservoir

Source for these calculations: iceclimate.h; surfaceclimate.h; planet.cpp.

Freezing requires removal of latent heat, so sea ice is represented by an energy reservoir rather than an immediate temperature switch. For column depth D, L_A is the latent heat per unit area, f is frozen mass fraction, and h is the equivalent ice thickness:

$$L_A = \rho_W D L_f, \quad f = C(-E/L_A; 0, 1), \quad h = fD\rho_W/\rho_I$$

E is the energy per unit area measured relative to an unfrozen column at its freezing point. The latent heat is L_f=334000 J kg⁻¹; water and ice densities are ρ_W=1000 and ρ_I=917 kg m⁻³. Energy below zero first represents the latent heat associated with freezing. Once the entire column is frozen, further energy loss cools the ice.

The default water-column and mixed-layer depth is 50 m. Ice conductivity is 2.2 W m⁻¹ K⁻¹. Those application choices accompany an enthalpy-based approach motivated by [R3].

Liquid and fully frozen columns have different effective heat capacities. C_W includes the water mixed layer of depth D_{mix}; C_I includes a capped thickness of ice together with atmospheric heat capacity C_A:

$$C_W = C_A + 4.18 \times 10^6 \min(D, D_{mix}), \quad C_I = C_A + 1.93 \times 10^6 \min(2, D\rho_W/\rho_I)$$

In the fully liquid regime, positive energy raises temperature above the freezing point. Once the column is fully frozen, energy below $-L_A$ cools the ice. $T_f(1)$ denotes the freezing reference used at the fully frozen endpoint:

$$T = T_f + E/C_W \quad (E \geq 0), \qquad T = T_f(1) + (E + L_A)/C_I \quad (E \leq -L_A)$$

Partially frozen water lies between the fully liquid and fully frozen regimes shown above. As ice forms, salt is rejected into the remaining liquid, increasing its salt-to-water ratio inversely with the remaining water fraction, up to 264 g kg^{-1}. That changes the local freezing temperature T_f.

The surface temperature in this intermediate regime follows a conductive balance and cannot exceed the applicable freezing temperature.

The same thickness h affects visible ice cover and the climate at different rates. The exponential c_{ice} controls optical whiteness, while a_{clim} sets the growth of the albedo effect:

$$c_{ice} = 1 - e^{-h/0.1}, \quad a_{clim} = \tanh(h/0.5)$$

Ice can become visibly white before it is thick enough to have its full climatic effect. The optical cover c_{ice} and the albedo-response factor a_{clim} therefore use different thickness scales. The water and ice albedos are .07 and .6, with default visibility .5.

Changing albedo changes absorbed shortwave radiation. It does not remove transported energy: ocean heat transport is divided between the ice base and atmosphere while preserving its total.

The energy step is written implicitly: the net energy flux F_{net} is evaluated at the new state E_{new}. Over interval Δt, the change in stored energy must equal that flux integrated over the interval:

$$E_{new} - E_{old} = \Delta t F_{net}(E_{new})$$

The implicit step solves for the new energy using the energy flux at that new state. It can therefore cross freezing and melting boundaries without treating phase changes as instantaneous switches.

Periodic histories use 17 rows spaced in sine latitude and 128 orbital phases, with longitude resolution for synchronous worlds. Diagnostics report how closely the end of a cycle closes onto its beginning. A separate thin-frost approximation uses exchange of 10 W m^{-2} K^{-1}, 10 cm of ice for water or 1 cm for land, and a ±0.15 K dead band. That fallback is distinct from the finite-depth ocean energy reservoir.

20. Habitability flags, vegetation habitat, and radiation belts

Source for these calculations: planet.cpp; vegetation.h; habitablezone.h.

A habitable-zone boundary expresses the stellar flux at which a particular one-dimensional climate criterion is met. The polynomial uses coefficients S_0, a, b, c, and d together with a host-temperature offset ΔT, after limiting the stellar temperature to the fitted range:

$$S_{eff} = S_0 + a\Delta T + b\Delta T^2 + c\Delta T^3 + d\Delta T^4, \quad \Delta T = C(T_*; 2600, 7200) - 5780$$

The conservative habitable-zone boundaries use the Kopparapu runaway- and maximum-greenhouse relations [R9]. The polynomial estimates the effective stellar flux S_{eff} at a boundary as a function of the host's temperature offset ΔT.

For the inner boundary, coefficients are interpolated linearly between 0.1, 1, and 5 Earth masses. The outer coefficients are independent of planet mass, and the nearest endpoint approximation applies outside the inner-boundary mass range. These boundaries describe a one-dimensional water-based climate criterion, rather than evidence of life.

Boundary / mass	S_0	a, b, c, d
Inner / 0.1	.99	1.209e−4, 1.404e−8, −7.418e−12, −1.713e−15
Inner / 1	1.107	1.332e−4, 1.580e−8, −8.308e−12, −1.931e−15
Inner / 5	1.188	1.433e−4, 1.707e−8, −8.968e−12, −2.084e−15
Outer	.356	6.171e−5, 1.698e−9, −3.198e−12, −5.575e−16

The older global surface-habitability flag combines several explicit tests: location within the orbital zone, mass above .02 Earth masses, waterline at least .05, pressure from 600 Pa to below 2000 atmospheres, and mean temperature from .9×273.15 K to 320 K.

Passing those tests is only a classification result. A waterline threshold is not a minimum measured area of exposed liquid, and the lower temperature gate of approximately 245.8 K does not alone establish liquid water.

Moons of giant planets can be assigned reduced surface-life suitability near an assumed radiation belt. The distance-dependent score P_{belt} rises through transition distance d_0, with slope s:

$$P_{belt}(d) = 0.01 + \frac{0.99}{1 + (d_0/d)^s}$$

For a moon orbiting a gaseous planet, d is distance in units of the parent's equatorial radius. The transition distance d_0 starts at 8+12U and is multiplied by mass and spin factors. These are (M_{parent}/M_J)^(1/6), clipped to [.6,1.8], and (10 h/$|P_{spin}|$)^(1/6), clipped to [.7,1.4]. The slope is s=3+2U.

The model averages the score around the orbit using 32 eccentric-anomaly samples with duration weights 1−e cos E.

An atmospheric column can provide shielding in this appearance scenario. The ratio (p/g)/(101325/9.80665) compares its mass per unit area with Earth's, reducing the penalty associated with the orbit-averaged belt score:

$$P_{shield} = 1 - (1 - \langle P_{belt} \rangle)\exp[-(p/g)/(101325/9.80665)]$$

A repeatable draw tied to object identity determines whether simulated surface life survives this rule. The score represents an assumed radiation-belt environment, not a magnetic-field solution, particle spectrum, radiation dose in grays, or measured biological survival probability. It also contributes no climate heating.

The atmospheric column mass p/g supplies a heuristic shielding effect. Other kinds of orbital parent receive a score of one under this particular rule.

21. Vegetation adaptation and photosynthetic reflectance

Source for these calculations: vegetation.h; color.cpp.

The vegetation scenario adapts a temperature interval to the modeled climate. The random adaptation factor a_v sets how strongly its center T_c follows the survey's cold and hot temperatures instead of the 294 K reference:

$$a_v = 0.55 + 0.30U, \quad T_c = 294 + a_v[(T_{cold} + T_{hot})/2 - 294] + U(-8, 8)$$

The half-width ΔT_v follows the surveyed temperature spread, with additional random variation. Clipping keeps that half-width between 12 and 60 K:

$$\Delta T_v = C\{[16 + a_v((T_{hot} - T_{cold})/2 - 16)](0.65 + 0.45U); 12, 60\}$$

The vegetation model first surveys the climate at 13 sine-latitudes and 24 longitudes, both at sea level and at representative upland elevations. It uses the sampled cold and hot conditions to choose an adapted temperature center T_c and half-width ΔT_v. The final limits T_c±ΔT_v are clipped to 223.15–453.15 K.

Those broad biochemical limits deliberately permit speculative vegetation; they are not measured universal growth limits. Habitat is then checked through the year and day using 33 latitude rows, 16 longitude columns for synchronous worlds, and eight daily phases otherwise. At least one tenth of the samples must lie within the adapted temperature range, the liquid-water window, and the biochemical bounds.

Stellar spectrum also influences the assumed pigment bands. Multiplying Planck radiance by wavelength gives a relative photon-supply weighting, with the constant photon-energy conversion cancelling in the normalization. The added floor leaves every sampled wavelength a nonzero chance:

$$w_\lambda = 0.20 + 0.80\frac{\lambda B_\lambda(T_*)}{\max(\lambda B_\lambda(T_*))}$$

The probability weight w_λ favors wavelengths with a larger available photon supply from the star. Pigment centers are drawn from 350–1000 nm at 5 nm intervals, then shifted by up to ±2.5 nm. At least one center lies in the visible interval, 400–700 nm.

Each spectrum has two through five absorption bands. The first two widths are U(10,25) nm, and accessory widths are U(20,45) nm. A common concentration is drawn log-uniformly from .6 to 5, and each band depth receives an additional U(.5,1.5) factor.

Each pigment absorption band is a Gaussian centered at λ_j, with width s_j and depth d_j. Their sum gives pigment optical depth $\tau_p(\lambda)$:

$$\tau_p(\lambda) = \sum_j d_j \exp\left[-\frac{(\lambda-\lambda_j)^2}{2s_j^2}\right]$$

The hypothetical reflectance spectrum R(λ) combines a visible-to-infrared scattering background with pigment absorption and a UV cutoff. λ_e and w_e locate and broaden the structural edge, while λ_{UV} sets the ultraviolet transition:

$$R(\lambda) = C\left[\left(r_V + \frac{r_{IR}-r_V}{1+e^{(\lambda_e-\lambda)/w_e}}\right)\frac{e^{-\tau_p(\lambda)}}{1+e^{(\lambda_{UV}-\lambda)/10}}; 0, 1\right]$$

The unabsorbed background has visible scattering r_v=U(.15,.65) and infrared scattering min(.90,r_v+.25). Its UV edge lies at U(320,380) nm, and its structural reflectance edge is max(700,max_j(λ_j+2s_j)). Together with the absorption bands, these choices make a hypothetical leaf-like reflectance spectrum.

This does not identify an alien pigment or calculate oxygenic photosynthetic efficiency. The display RGB values average reflectance over the overlapping windows 560–709, 480–599, and 400–509 nm. Absorption can therefore leave a dark spectrum; the result is not automatically normalized to unit brightness.

A separate appearance preference favors purple over green by 5:2 among sufficiently saturated candidates in those hue families. The generator can try up to 128 pigment mixtures; other hue families are accepted without that preference. This is an artistic population choice. It carries no empirical claim about extraterrestrial ecology, and stellar illumination is applied separately from intrinsic reflectance.

22. Gradual forest margins

Source for these calculations: vegetation.h; surfacenavigation.cpp; celestial.cpp; visuals.cpp.

A forest margin should become sparse as conditions become marginal. The habitat model therefore assigns a continuous density rather than using only a yes-or-no temperature test.

The allowed interval [L,H] is the overlap of the adapted growth range, the liquid-water interval, and the speculative biochemical limits. A margin width w=min(12 K,(H−L)/4) controls how gradually suitability declines near each endpoint. The clipped cubic smoothstep S maps its argument into [0,1] to produce the thermal score q_T.

The local thermal score q_T tapers within the allowed interval. Averaging it through the sampled climate cycle and applying a second smooth transition gives the thermal density D_T:

$$q_T(T) = S[(T-L)/w]S[(H-T)/w], \quad D_T = S[(\langle q_T \rangle - 0.1)/0.3]$$

The average of q_T samples the year and day, including altitude and regional temperature offsets. A short or marginal growing season produces a low density D_T; consistently suitable conditions approach full density. The older binary suitability diagnostic remains available, but the visible forest uses this continuous score.

Local evaluation interpolates terrain elevation between map samples. Seasonal habitat tables are retained until their physical inputs change.

On a star-locked world, persistent darkness can reduce habitat even where temperature is suitable. The solar projection n·s supplies an additional light factor D_L, multiplied by D_T to give the final density D:

$$D_L = S[(\mathbf{n}\cdot\mathbf{s})/0.20], \quad D = D_T D_L$$

The illumination factor applies where the same terrain remains near the star-facing hemisphere of a locked world. It rises from zero at the terminator to full density when the solar projection reaches .20. On rotating worlds this particular factor is D_L=1, so an ordinary night does not remove a mature forest.

The width of the transition is an ecological appearance assumption. It is not a measured universal light-compensation point for photosynthesis.

That continuous density controls two visible decisions. A site-specific draw determines whether a whole tree is present, and the ground palette interpolates between vegetation and bare-ground colors:

$$\text{tree present} = \mathbf{1}_{U_{site} < D}, \quad C_{ground} = DC_{vegetation} + (1-D)C_{bare}$$

Each geographic site receives a repeatable number U_{site}. Comparing it with density D removes whole trees as habitat becomes poorer, producing a sparse forest rather than a dense translucent one. Geographic identity keeps that choice independent of the camera.

When a procedural terrain map is rebuilt, its ground color also blends toward neighboring bare ground as D declines. Individual crowns retain the vegetation reflectance; lighting and

haze act afterward. Existing maps still receive continuous local tree thinning even before such a rebuild.

23. Trunks, crown dimensions, foliage texture, and placement

Source for these calculations: plantform.h; plantmesh.h; planttexture.h; visuals.cpp; surfacescatter.h.

Default plant dimensions respond to two bounded environmental factors. g_v is gravity relative to Earth, and p_v is a logarithmic pressure factor normalized to one at terrestrial pressure:

$$g_v = C(g/g_\oplus; 0.1, 10), \quad p_v = C\left[\frac{\ln(1+\max(0, p/p_\oplus))}{\ln 2}; 0.25, 4\right]$$

These factors set trunk height H_t, trunk width W_t, crown radius R_c, and crown shape ratio o_c. Their explicit bounds keep the generated geometry finite:

$$H_t = 10/g_v, \quad W_t = 0.3g_v, \quad R_c = \min(2.5/p_v, 10), \quad o_c = 0.8p_v$$

H_t and W_t set trunk height and width, R_c sets horizontal crown radius, and o_c is the vertical-to-horizontal crown ratio. They act as multipliers on the local scenery size. Crowns are spheroids, supported by capped trunks.

Crown color comes from the world’s vegetation reflectance at every detail level, independently of the terrain pixel beneath it. A tree over rock-colored habitat therefore retains its foliage pigment, while crown texture, lighting, and haze can still alter the displayed color.

Supplied nonzero morphology settings replace these defaults. An absent or nonfinite environmental scale falls back to one. The original audit replaced ln(p/p⊕)+1, which became zero at p=p⊕/e and negative below it, with the positive bounded pressure factor shown here. Bounding gravity also keeps trunk dimensions finite at low g.

The inverse-gravity trend gives taller trunks where structural support is easier in a visual analogy. The pressure trend gives broad, flat crowns at low pressure and narrower, taller crowns at high pressure.

These shapes are assumptions for scenery. The model does not solve tissue strength, wind loading, fluid transport, competition, or evolution, so the dimensions should not be interpreted as predictions of alien biomechanics.

Within those overall dimensions, foliage texture uses the following ranges. Cluster and leaf scales are lengths, while density, contrast, roughness, and vertical stretch control the repeating volume pattern:

Foliage parameter	Generated range
Cluster scale	.25–1.4 m
Leaf scale	.025–.18 m
Density	.35–.95
Contrast	.30–.75
Roughness	.08–.40
Vertical stretch	.45–2.8
Pattern	Four equally indexed pattern families

Foliage is represented by a three-dimensional repeating density field. Cluster and leaf lengths set its large and small features. Interpolation uses cubic smoothstep weights; the analytic gradient of the same field supplies surface shading consistent with its opacity.

Fine structure fades as its projected size falls below the viewing footprint. The chosen lengths are appearance parameters, rather than dimensions derived from stellar spectrum or atmospheric chemistry.

For a point x inside a lattice cell, the foliage value N(x) combines the eight corner values v_c. A product of three smooth weights interpolates across the cell, one weight for each coordinate:

$$N(\mathbf{x}) = \sum_{c \in \{0,1\}^3} v_c \prod_{k=1}^{3} w_{c_k}(x_k), \quad w_1(t) = 3t^2 - 2t^3, \quad w_0 = 1 - w_1$$

Scenery candidates are tied to geographic cells, with terrain material and habitat determining what can occupy a site. A crown-radius rejection test removes overlapping large crowns. Distance and projected detail then determine which accepted objects are drawn; changing those drawing choices does not change the underlying life-suitability calculation.

24. Hydrostatic scale height and cloud condensation

Source for these calculations: cloudclimate.h; planet.cpp.

The atmospheric column first supplies the conditions in which clouds can condense. Mole fractions x_i and molar masses μ_i give the mean molar mass $\bar{\mu}$, then the specific gas constant R_s. In an isothermal ideal-gas approximation, H is pressure scale height and m_{col} is column mass:

$$\bar{\mu} = \frac{\sum_i x_i \mu_i}{\sum_i x_i}, \quad R_s = R_g/\bar{\mu}, \quad H = R_s T/g, \quad m_{col} = p/g$$

The gas fractions x_i are normalized locally for this calculation while the supplied composition itself is retained. Approximate ideal-gas molar heat capacities are H_2 28.8, He/Ar 20.8, N_2 29.1, O_2 29.4, CO_2 37.1, H_2O 33.6, CH_4 35.7, and NH_3 35.1 J mol^{-1} K^{-1}.

When no mixture is available, the model uses approximate air values: molar mass .02897 kg mol^{-1} and molar heat capacity 29.1 J mol^{-1} K^{-1}.

The model samples 64 upward steps in log pressure. Each step reduces pressure from surface value p_s and estimates cooling through exponent κ, while a skin-temperature floor prevents indefinite extrapolation:

$$p_j = p_s e^{-0.12j}, \quad T_j = \max(T_{skin}, T_{j-1} e^{-0.12\kappa}) \quad (j = 1, \ldots, 64)$$

When water is saturated, condensation partly offsets cooling. The moist correction includes latent heat L, moisture loading q, water-vapor gas constant R_v, and the mixture heat capacities per mole and per unit mass:

$$\kappa = \frac{R_g}{c_{p,mol}} \frac{1 + Lq/(R_s T)}{1 + L^2 q/(c_{p,s} R_v T^2)} \quad \text{(saturated water)}$$

Unsaturated levels use $\kappa = R_g/c_{p,mol}$. Where water is saturated between 235 and 647 K, the moist correction uses latent heat $L = 2.5 \times 10^6$ J kg^{-1} and water-vapor gas constant $R_v = 461.5$ J kg^{-1} K^{-1}. The quantities q and $c_{p,s}$ represent the moisture loading and specific heat capacity per unit mass in that correction.

The upper-temperature floor is the smaller of surface temperature and max(30,.72T_eq scaled to the local temperature bin). Seventeen surface-temperature bins cover .55–1.25 of the mean. This vertical profile estimates condensation conditions without solving a radiative-convective weather system.

At each level, the vapor partial pressure x_i p_j is compared with saturation pressure $p_{sat,i}$. The largest weighted excess through the column defines b_i, a dimensionless measure of condensate availability:

$$b_i = \max_j \left[e^{-0.12j} \max\left(0, 1 - \frac{p_{sat,i}(T_j)}{x_i p_j}\right) \right]$$

The availability factor b_i measures how much of a species could condense along the estimated vertical profile. It supplies the column paths for water or ice, CO_2, methane, and ammonia.

If humidity is absent and an ocean exists without a specified gas mixture, the model assumes relative humidity $.75\sqrt{f_o}$ for warm water or $.20\sqrt{f_o}$ for ice. The inferred vapor mole fraction is capped at .2. A specified dry mixture remains dry rather than receiving this fallback.

25. Cloud optical depth, cover, and condensate colors

Source for these calculations: cloudclimate.h; cloudlighting.h; celestial.cpp.

Cloud darkness and opacity depend on condensate mass as well as coverage. The first relation converts condensate path W_c to optical depth τ_c using particle density ρ_c and effective radius r_{eff}. The second estimates a species column from the atmospheric inventory and assumed efficiencies:

$$\tau_c = \frac{3W_c}{2\rho_c r_{eff}}, \qquad W_i = \frac{p}{g}\frac{x_i\mu_i}{\bar{\mu}} b_i \varepsilon_i \zeta$$

W_c is condensate mass per unit area, in kg m^{-2}. For particles large compared with the wavelength, an extinction efficiency near two leads to the optical-depth expression above. Smaller effective radius r_{eff} gives more extinction for the same mass.

For species i, the column path W_i combines atmospheric column mass, condensable mass fraction, availability b_i, precipitation efficiency ε_i, and settling factor ζ. The latter is clip(√(9.80665/g),.2,3)×p/(p+1000). Particle sizes, efficiencies, and cover fractions are assumed microphysical settings rather than observations.

Condensate	Radius / density	Efficiency and coverage
Water liquid	10+5f_o µm / 1000 kg m^{-3}	ε=.0015; coverage clipped from .12+.72√RH+.08f_o
Water ice	20 µm / 917 kg m^{-3}	ε=.02; blends with water across 273.15–235 K
CO_2	30 µm / 1600 kg m^{-3}	$\varepsilon=10^{-5}$; coverage .18
Methane	20 µm / 500 kg m^{-3}	ε=.0005; coverage .12
Ammonia	15 µm / 700 kg m^{-3}	ε=.0005; coverage .25
Sulfuric acid	1 µm / 1800 kg m^{-3}	precursor conversion 10^{-4}; coverage .98

Methane saturation is anchored at 111.66 K and one atmosphere. It uses latent heats 9140/8200 J mol^{-1} and critical temperature 190.56 K. Ammonia is anchored at 239.82 K and one atmosphere, with latent heats 29000/23350 J mol^{-1} and critical temperature 405.4 K. The assumed phase switches are 90.69 K for methane and 195.4 K for ammonia.

The model sums optical depths, capped at 100, while taking coverage from the largest contributing layer. The species providing the greatest opacity supplies the condensate label.

An empirical convection factor helps set cloud organization. It rises with temperature above 240 K, relative humidity RH, and the pressure factor p/(p+3000):

$$c_v = C((T-240)/50;0,1)\sqrt{C(RH;0,1)}\frac{p}{p+3000}$$

The convection proxy c_v above grows with warmth, relative humidity RH, and pressure. It helps set the amount of organized cloud structure in the later appearance calculation.

A warm sulfuric-acid layer is permitted if the vertical profile crosses 200–400 K, surface temperature exceeds 300 K, and the mixture contains SO_2, H_2O, and either CO_2 or O_2. Its available precursor is $\min(x_{SO_2}, x_{H_2O})$ times .098079/μ. This is a sulfate-production proxy rather than a photochemical reaction network.

Water, ice, and other non-acid decks use bright RGB (244,246,249) and deep RGB (195,199,205). Acid decks draw warmer bright values from (242–249,234–243,214–228), with the deep channels reduced by (.78–.86,.66–.75,.44–.56).

These values prescribe a display palette. They are not measured refractive indices. Stellar color, illumination, shadows, and viewing depth continue to change the rendered result.

26. Rocky-world cloud shapes and storms

Source for these calculations: celestial.cpp: cloud-map generation; clouds_gpu.cpp.

The cloud map uses rotation to choose a degree of organization. □ compares rotational speed ωR with an atmospheric thermal-speed scale √(R_s T). The activity factor A combines that rotation proxy with convection c_v and optical thickness τ_c:

$$\mathcal{R} = C\left[\frac{\omega R}{1.7\sqrt{R_s T}};0,3\right], \quad A = c_v \min(1,\tau_c/3) C(\mathcal{R};0,1)$$

The activity factor A scales an integer draw from 7 through 10 to obtain the storm count, rounded to the nearest integer. One in four candidate systems is compact; the others have broader frontal heads. Their initial angular radii are U(.08,.13) and U(.14,.24) radians, multiplied by clip(1/√(□+.15),.6,2.5).

Compact systems occupy absolute latitudes .20–.52 rad and broad systems .44–1.09 rad. Hemisphere and spin direction determine their winding. Up to 48 placement attempts keep storm cores apart.

A storm is described in coordinates centered on its direction c. The angle from surface direction n to c, divided by angular radius r_{storm}, gives normalized distance r. A signed winding factor s then supplies the twist angle:

$$r = \frac{\arccos(\mathbf{n}\cdot\mathbf{c})}{r_{storm}}, \quad \theta_{twist} = s(5+3U)(1-r)^2$$

The storm-centered east/north coordinates are rotated by θ_{twist}, with an envelope that fades between normalized radii .72 and 1. Fronts are bent using distance u−b(v^2−.15), with b=U(.35,.80).

Compact storms combine a Gaussian cloud shield with an eye cleared over .035–.11 of the normalized radius. Broad storms combine a head and a curved frontal tail. These shapes illustrate organized weather without integrating pressure, wind, or the Navier–Stokes equations.

Several noise scales fill the displaced storm coordinates w. N_o denotes the multiscale noise defined in the terrain section, with subscript giving octave count. Broad, detailed, and elongated fields are evaluated separately:

$$F_b = N_5(4\mathbf{w}), \quad F_d = N_4(18\mathbf{w}), \quad F_f = N_4(9w_x, 9w_y, 36w_z)$$

The coordinate w follows the sphere after storm displacement and an additional low-frequency warp of amplitude .35. The noise fields F_b, F_d, and F_f set broad bodies, small detail, and elongated fine features respectively.

Cloud bodies use smooth thresholds .48–.57 plus .22(detail−.5). Wisps combine thresholds .42–.50 and .49–.60. Optical columns vary on scales 2.6 and 30. Coverage shifts the thresholds by .18×(coverage−.65), while clip(τ_c/6,0,4) sets the optical scale. The finest noise octave is limited to about two pixels.

Cloud opacity increases with optical depth rather than linearly with thickness. The body and wisp masks b and f are multiplied by exponential transmission complements; d_b and d_f vary their depths and τ_* is the overall optical scale:

$$\alpha_{body} = b(1 - e^{-d_b \tau_*}), \quad \alpha_{wisp} = 0.65f(1 - e^{-d_f \tau_*})$$

The depth factors give body depth (.55+3.80column)(.55+.80cells), wisp depth (.12+.65column)(.80+.40cells), and storm depth (1.20+3.20column+1.50shape)(.70+.60cells). Here “column,” “cells,” and “shape” are local procedural fields used to vary thickness. The final opacity is the largest of the storm, background, and overcast-deck contributions.

A map is classified as globally opaque only if every pixel reaches opacity 179/255. Its weather random state combines object identity, simulation hour, and weather revision. Changing the epoch can therefore generate another weather realization, rather than advancing a continuous forecast.

27. Gaseous-world bands, vortices, and overcast decks

Source for these calculations: giantclouds.h; celestial.cpp.

A giant planet’s cloud map begins with bands organized by latitude. The phase ψ combines latitude coordinate z, band frequency ν, phase offset φ, and broad and fine noise fields N_b and N_f. The smooth band weight B then selects between palette regions:

$$\psi = z\nu + \phi + 1.4N_b + 1.2N_f, \quad B = S\left(\frac{0.5 - 0.5\cos\psi - 0.18}{0.62}\right)$$

Spherical noise and latitude-dependent shear bend the boundaries of the bands. Broad structure uses scales 3.2 and 4.7; tracer detail starts at scale 12, with longitudinal shear .12 sin(zν+φ). Three-octave folds and threads add finer features.

As angular pixel size or local vortex strain increases, the smallest structures fade. This limits aliasing while retaining the larger cloud pattern.

To place a vortex in that band field, the model measures elliptical distance r from vortex center (λ_v,φ_v). Longitude separation is shortened by cos φ_v, while w_v and h_v set its angular width and height:

$$r^2 = \left[\frac{(\lambda - \lambda_v)\cos\varphi_v}{w_v}\right]^2 + \left[\frac{\varphi - \varphi_v}{h_v}\right]^2$$

Within normalized radius 2.4, a bounded twist rotates the sampling coordinates. t_v controls the winding and the noise term N varies its strength locally:

$$\theta = t_v(1 - r^2/2.4^2)^3(1 + 0.23N) \quad (r < 2.4)$$

The vortex transformation displaces the bands and their tracers together, so a storm appears embedded in the flow. A major storm has height U(.075,.12) rad, aspect ratio U(2.7,3.8), and twist magnitude U(3.5,5).

Smaller vortices have latitude U(−1.14,1.14) rad, height U(.014,.047), aspect ratio U(1.4,3.1), and twist U(1.7,3.5). A 40% draw selects a bright variant. Banded patterns contain 38 small vortices and hazy patterns seven. An eligible non-synchronous world receives a major storm with a 20% draw.

The band coordinate selects interpolated colors from a palette after displacement by broad noise, folds, and threads. Fine fields alter brightness while retaining the local hue, and blending toward the poles reduces contrast.

The clear style returns the base color. For synchronous worlds, longitude is sheared by jet strength×$\cos^{15}$(latitude) and then expressed about the star-facing axis. The bands consequently bend around the illumination geometry. These are static reflectance patterns, not a solved circulation.

A separate opaque-deck texture blends between deep and bright colors. Its interpolation weight t combines a polar term |z| raised to k_p, contrast c, broad noise N, and smaller-scale modulation mN_m:

$$t = C[|z|^{k_p} + c(N - 0.5) + mN_m; 0, 1], \quad C_{deck} = (1 - t)C_{deep} + tC_{bright}$$

The overcast-deck model draws zonal stretching U(5,9), spatial scale U(1.6,2.6), shear U(1.5,4), exponent U(1,1.6), warp amplitude U(.5,1.1), polar exponent U(1.8,3.6), and contrast U(.38,.62). Rotation changes the stretching, and increasing optical thickness reduces contrast.

For a synchronous deck, the star-axis cosine replaces poleward latitude in the blend. The shear parameters organize appearance and do not estimate actual atmospheric jet speeds.

28. Giant reflectance spectra and band colors

Source for these calculations: giantreflectance.cpp; giant_reflectance.json.

Gas- and ice-giant reflectance can be drawn from the Cahoy, Marley, and Fortney library [R10]. The stored grid has 16 models: two atmosphere families, two enrichment values in each family, and distances of .8, 2, 5, and 10 AU. Spectra are sampled every 5 nm from 350 to 1000 nm.

Within a family, distance and enrichment are interpolated logarithmically. Mass controls the mixture between the families. Values outside the grid are held at an endpoint, and the diagnostic record reports that limit.

The atmosphere-family weight w_N moves from a Jupiter-like family at 318 Earth masses toward a Neptune-like family at 17 Earth masses. In this section m is planetary mass expressed in Earth units:

$$w_N = C\left[\frac{\ln(318/m)}{\ln(318/17)}; 0, 1\right]$$

The enrichment Z gives the atmospheric heavy-element scaling within the assumed family. It depends on mass, stellar [Fe/H], and a random offset, and is limited to the available scenario range:

$$Z = C[3(318/m)^{0.6}10^{0.25[Fe/H] + 0.3(U - 0.5)}; 1, 30]$$

Each visible band samples a perturbed equivalent orbital distance a_{band}. The base value a_{eq} is adjusted weakly for surface gravity g and by two independent random offsets, representing cloud variation within the library-based scenario:

$$a_{band} = a_{eq}(g/15)^{0.04}10^{0.12(U_1 - 0.5) + 0.16(U_2 - 0.5)}$$

Z represents the enrichment of the assumed atmosphere. It is a scenario setting rather than a measured bulk metallicity. Each of eight band realizations adds cloud fraction U(.04,.16) and a blue-absorber coefficient h×U(.3,1.7). Unless supplied, h is drawn from U(.08,.38).

The equivalent distance from the Sun is $1/\sqrt{F}$ for irradiation F in Earth units. Where a temperature proxy supplies the estimate, it is $(255/T_{proxy})^2$.

The band reflectance mixes a library spectrum $R_g(\lambda)$ with cloud reflectance 0.65 at fraction c. A wavelength-dependent exponential then absorbs preferentially in the blue. h_b is its band-specific strength, and λ is expressed in nanometres:

$$R_{band}(\lambda) = [(1-c)R_g(\lambda) + 0.65c]\exp[-h_b(450/\lambda)^4]$$

At wavelengths λ≥600 nm, a local maximum within ±60 nm provides a continuum estimate. Raising trough transmission to a methane-dependent power changes the absorption depth relative to that continuum.

A supplied methane fraction sets the scale through $\text{clip}(x_{CH_4}/(.0005Z),0,4)$. Otherwise the scale comes from a bounded power of the carbon-to-iron abundance ratio. The procedure adjusts a library spectrum; it does not recompute atmospheric chemical equilibrium.

Converting that spectrum to a displayed color requires an illuminant and a color response. The sum weights reference solar flux F_⊙ by the host's Planck-spectrum ratio, reflectance R, integration weights w_λ, and CIE matching functions $\bar{x}$_k to obtain each tristimulus coordinate X_k:

$$X_k \propto \sum_\lambda w_\lambda F_\odot(\lambda)\frac{B_\lambda(T_*)}{B_\lambda(5778)}R(\lambda)\bar{x}_k(\lambda)$$

The CIE (International Commission on Illumination) color-matching functions describe a standard human observer's response to wavelength. Trapezoidal integration with these functions gives the XYZ color coordinates. A standard linear RGB conversion follows, with normalization to the peak channel and clipping to the display gamut. Separately, B, V, and R filter integrals relative to the reference illuminant give photometric colors.

The library describes reflection at opposition. Normalizing its display RGB retains hue but removes absolute reflectivity, so those normalized channels alone cannot supply the object's physical brightness. That information comes from the photometric model.

29. Terrain noise, rocks, craters, and molten surfaces

Source for these calculations: misc.cpp; noise.cpp; celestial.cpp; rockmeshcache.h; lavafissures.h.

Terrain begins as a smooth spatial noise field rather than a list of explicit landforms. Successive octaves add finer structure: n is the elementary noise, o the octave count, ℓ the frequency ratio, and g the amplitude gain. Here g is a noise parameter, distinct from gravitational acceleration:

$$N_o(\mathbf{x}) = \frac{1}{2}\left[1 + \frac{\sum_{j=0}^{o-1} g^j n(\ell^j \mathbf{x})}{\sum_{j=0}^{o-1} g^j}\right]$$

A related field folds the noise into ridges before combining the scales. Q_o therefore has sharper crests than the ordinary multiscale field N_o:

$$Q_o(\mathbf{x}) = \frac{\sum_j g^j [1 - |n(\ell^j \mathbf{x})|]^2}{\sum_j g^j}$$

The elementary gradient noise n uses lattice gradients, a shuffled 256-entry permutation, and the quintic fade $6t^5-15t^4+10t^3$. Adding successively finer scales gives N_o; folding the noise around zero gives the ridged field Q_o.

Frequencies, gains, coordinate offsets, and warps set a visual distribution of features. They are not measured geological power spectra. Evaluating the fields on a sphere keeps longitude continuous at the map seam.

Rocky terrain combines five through eight octaves. The feature scale is U(1.5,2.9) with water and U(.2,.8) without it; the corresponding gains are .5 and 2.5. The frequency ratio is U(.51,.53)√max(1,ln(R expressed in metres)). The metre reference is part of the visual calibration.

Coordinate warping and regional roughness change local gain. Separate spherical fields add color provinces, grain, mottled boundaries, and red/blue hue variation. Organic staining is restricted to cold thermal regimes, but the resulting colors do not identify particular minerals.

Crater count increases with bombardment enhancement B and decreases with pressure p relative to Earth. The separate diameter draw D spans 50 m to the smaller of 0.3R and 900000 m, with U^5 favoring the small end:

$$N_{crater} = \left\lfloor N_0 \frac{B}{1 + 3p/p_\oplus} \right\rfloor, \quad D = 50 + [\min(0.3R, 900000) - 50]U^5$$

The reference count N_0 is 300 for ordinary surfaces and three for molten surfaces. Crater centers are sampled uniformly by area. Depth is the visual-relief scale multiplied by U(.3,.9), with rim height .15–.35 of that depth and rim width .15–.30 of angular radius.

Large basins may have central peaks, and rare rays use angular lobes. The U^5 diameter draw favors small craters. It supplies a synthetic size distribution rather than a calibrated record of impact age.

Local boulder shapes use an assumed pressure-dependent slab preference b_{slab}. It reduces the rock aspect parameter a_{rock} from its material-based scale a_m, with clipping to keep the geometry within the chosen range:

$$b_{slab} = C[\log_{10}(p/p_\oplus)/3; 0, 1], \quad a_{rock} = C[a_m(0.45 + 1.35U)(1 - 0.68b_{slab}); 0.045, 2.2]$$

Each boulder starts from a perturbed 26-vertex dome. Horizontal dimensions are scaled by U(.70,1.30) and U(.65,1.30), while vertical scale is U(.85,1.15). Small tilts and an offset apex avoid regular symmetry.

Three profile families give broad caps, unequal shoulders, or taller crests. The pressure-dependent slab preference changes the appearance distribution; it does not calculate erosion.

Molten surfaces combine warped six-octave terrain, fissures, trigonometric vent patterns, and low basins. Fissures lower the normalized terrain by .08; vents activate above .64 in their sine/cosine field.

Piecewise red, orange, and gold color ramps represent activity and are multiplied by a thermal-emission factor. Reflected basalt remains visible in daylight. The palette illustrates molten material without serving as a calibrated emission spectrum or a model of magma viscosity and convection.

30. Sky colors, extinction, yellowing, and dense haze

Source for these calculations: visuals.cpp; color.cpp; pressurehaze.h.

For the broad sky-color estimate, C_{sky} is a three-channel color vector. A particulate fraction f_d interpolates between a fixed blue scattering color and the planetary surface color C_{planet}:

$$\mathbf{C}_{sky} \propto (1 - f_d)(0.37, 0.58, 0.81) + f_d \mathbf{C}_{planet}$$

The sky starts from a blue vector chosen to resemble Rayleigh scattering. A particulate fraction f_d blends it with the planetary surface color, after which daylight, starlight, and optional city light modify its intensity.

Grazing paths use channel-dependent exponentials with a pressure-fourth-root depth bounded between .2 and 3. Apparent limb strength grows as min(1,(p/p⊕)^(1/4)). These are compact display approximations to scattering. A molten-world option instead supplies warm mineral-vapor colors.

Light near the horizon traverses a longer atmospheric path. The proxy X(z), with zenith angle z, increases the estimated color excess Δ(B−V); pressure p is referenced to terrestrial pressure:

$$X(z) = \frac{1}{\cos z + 0.025}, \qquad \Delta(B - V) = 0.2(0.14)\frac{p}{101325} X(z)$$

The yellowing calculation mirrors zenith angles beyond 90° and uses .025 in the denominator to keep the horizon expression finite. The terrestrial color excess of .14 mag per airmass and display multiplier .2 are empirical settings; some direct-light paths apply another factor of three.

The expression does not calculate composition-specific extinction or the true curved-atmosphere path length. At extreme pressures it can exaggerate color changes, so the result is a display adjustment rather than a measured color excess.

Dense haze uses a more explicit scattering coefficient. The ideal-gas number density $p/(k_B T)$ is multiplied by a reference cross-section, composition factor r_{scat}, and a smooth pressure-activation term to give β_{550}:

$$\beta_{550} = \frac{p}{k_B T}(4.6 \times 10^{-31}) r_{scat} S\left(\frac{p - 2 \times 10^5}{8 \times 10^5}\right)$$

The red, green, and blue coefficients β_c then use the Rayleigh λ^{-4} dependence at representative wavelengths λ_c. The particulate fraction f_d adds a wavelength-independent contribution in this approximation:

$$\beta_c = \beta_{550}[(550/\lambda_c)^4 + f_d], \quad \lambda_c = (650, 550, 450)\ \text{nm}$$

Dense pressure haze begins above 2×10^5 Pa and reaches full activation at 10^6 Pa. The factor r_{scat} averages approximate Rayleigh strengths for the gas mixture. Number density is estimated with the ideal-gas law.

At high density, real-gas refractivity and correlations between molecules can matter. They are omitted from this appearance approximation.

Over path length d, direct contrast decays exponentially and haze contributes airlight L_c. H is the atmospheric scale height, so β_{550H} also sets an approximate sky opacity α_{sky}:

$$\alpha_{sky} = 1 - e^{-\beta_{550}H}, \quad C_c(d) = C_c(0)e^{-\beta_c d} + L_c(1 - e^{-\beta_c d})$$

Redistributed diffuse light uses an approximate two-stream transmission $T_{diff,c}$. A final power transformation expresses that transmission on the display scale:

$$T_{diff,\,c} = (1 + 0.75\beta_c H)^{-1}, \quad T_{display,\,c} = T_{diff,\,c}^{1/2.2}$$

The diffuse two-stream estimate represents redistributed light before the haze airlight is normalized for display. A cap based on luminance .2126R+.7152G+.0722B limits haze brightness to 140 in display units.

This visible haze model is separate from the infrared optical-depth model. It does not, by itself, calculate an anti-greenhouse energy balance.

31. Atmospheric refraction and the local horizon

Source for these calculations: planet.cpp; surfacerefraction.h.

Atmospheric refraction shifts apparent altitude, especially near the horizon. The distant-sky approximation first expresses pressure as P_r in Earth units and compresses unusually large values to keep the empirical formula controlled:

$$P_r = p/101325, \qquad P_r \mapsto 5 + \log_{10}(P_r - 4) \quad (P_r > 5)$$

A pressure-dependent width k smooths true altitude h into an adjusted altitude h_c near the horizon. The logarithmic expression softens the low-altitude transition:

$$k = 0.5\max(1, P_r)^{1.3}, \quad h_c = -1 + k\ln[1 + e^{(h+1)/k}]$$

The refraction estimate R_{deg} then scales the empirical altitude correction by pressure and temperature. In this equation h_c and the cotangent argument are expressed in degrees:

$$R_{deg} = \frac{1.02}{60}\cot\left[h_c + \frac{10.3}{h_c + 5.11}\right]\frac{283}{T - 0.15}P_r$$

The empirical altitude correction uses degrees inside the cotangent. It is based on a regularized Saemundsson-style relation, added to the true altitude of the object. The shift becomes zero once the angular argument reaches 90° and is otherwise clipped to 0–5°.

Compressing pressure and broadening the low-altitude transition reduce excessive sticking near the horizon. The underlying terrestrial formula cannot reproduce arbitrary atmospheric ducts or mirages.

The local ground horizon uses another approximation. Refractivity ν=n−1 is scaled from a standard gas value by pressure and temperature. Its vertical variation gives ray curvature κ_{ray}, using mean molar mass μ and lapse rate Γ:

$$\nu = (n-1)_{std}\frac{p}{101325}\frac{273.15}{T}, \quad \kappa_{ray} = \frac{\nu}{(1+\nu)T}\left(\frac{\mu g}{R_g} - \Gamma\right)$$

Comparing ray curvature with planetary curvature gives an effective coefficient c. For observer height h_{eye} above the surface, it sets the estimated horizon distance d_H and dip angle δ:

$$c = \tfrac{1}{2}\max(0.01/R,\, 1/R - \kappa_{ray}), \quad d_H = \sqrt{h_{eye}/c}, \quad \delta = \tan^{-1}(2\sqrt{h_{eye}c})$$

The local horizon model estimates how a density-dependent refractive index bends near-ground rays. A supplied lapse rate is retained. Otherwise .0065g/9.81 K m^{-1} blends toward .7 K m^{-1} for a dense, hot atmosphere. The transitions span refractivity .0005–.015 and temperature 320–700 K.

This is an assumed lower-atmosphere behavior, rather than a universal lapse law. The curvature bound limits the effective-radius singularity to 100 geometric radii. Local ground geometry and the altitude correction for the distant sky remain separate approximations.

32. Photometry, surface light, cloud light, and visibility

Source for these calculations: planet.cpp; color.cpp; cloudlighting.h; visuals.cpp; sphere_impostor.cpp.

Photometry separates intrinsic brightness from distance. Absolute magnitude M refers to the standard distance of 10 parsecs; apparent magnitude m changes with distance d. Magnitude differences encode the flux ratio logarithmically. Here M denotes magnitude rather than the mass used in other sections:

$$m = M + 5\log_{10}(d/10\text{pc}), \qquad F_1/F_2 = 10^{-0.4(m_1 - m_2)}$$

A planet's visible fraction changes with phase angle α. For a diffusely reflecting Lambert sphere, the normalized phase function is:

$$\Phi_L(\alpha) = \frac{\sin\alpha + (\pi - \alpha)\cos\alpha}{\pi}$$

A second analytic phase law uses slope parameter G to combine two empirical falloffs. The label HG here denotes this weighted phase-function form:

$$\Phi_{HG} = (1-G)e^{-3.33\tan^{0.63}(\alpha/2)} + Ge^{-1.87\tan^{1.22}(\alpha/2)}$$

The phase angle α describes the relative directions of the star and observer as seen from the planet. Analytic phase functions describe how integrated brightness changes with that angle. Solid-surface slope estimates and an opposition-surge term modify the falloff, and clouds can add a Lambert-like contribution.

The final brightness also depends on reflectivity, radius, the source's brightness, and both relevant distances. These expressions avoid solving a separate bidirectional reflectance function for every piece of terrain.

The main reflected-flux scaling can be summarized with geometric albedo A_g, planet radius R, phase function Φ, star–planet distance r_*, and planet–observer distance d_{obs}:

$$F_{refl} \propto A_g R^2 \Phi(\alpha)/(r_*^2 d_{obs}^2)$$

A surface's intrinsic reflectance is distinct from the spectrum illuminating it. The host spectrum is approximated through blackbody or color-index transformations. Pressure yellowing, ambient illumination, cloud shadows, and haze then alter the light reaching the observer.

Local scenery combines diffuse light with the positive surface-normal projection of direct light. Dense haze shifts some direct illumination into diffuse ambient light. Cloud shading also depends on optical depth, solar angle, and approximate forward and diffuse scattering. A texture's opacity does not measure cloud height.

The displayed result also depends on brightness, gamma, red-light mode, glare, bloom, exposure, and angular pixel size. A screen RGB sample consequently cannot determine a unique composition or surface reflectance.

Fine noise and distant meshes are filtered or simplified to suit their projected size. Such a sampling change alters what can be resolved, rather than the object's physical state.

33. Eclipse corona filaments

Source for these calculations: visuals.cpp: corona rendering.

A synthetic corona becomes visible near total obscuration of the stellar disk. Its amplitude is A=clip((obscuration−.97)/.03,0,1)^1.5, with values below .02 omitted. The bright inner halo extends from 1.01 to 2.2 stellar radii, with opacity 150A and radial falloff power 2.6. The faint outer halo extends from 1.01 to 5 radii, with opacity 28A and falloff power 1.7.

These settings represent display exposure. They do not infer the corona's absolute density. Curved filaments are then laid over the halos using the geometry below.

Each coronal strand follows a curved polar path about the star. Normalized coordinate t runs outward, R is stellar radius, and L is the strand's extent. The random values u_i keep neighboring lengths varied but repeatable:

$$r(t) = 0.985R + Lt, \quad L = R(0.65 + 3.8u_1^2)[0.8 + 0.2\cos(2\theta_0 - 0.7)]$$

The angular path starts at θ_0. Bend b gives a gradual turn, strand offset s separates the neighbors, and the final sinusoid adds a small outward-growing deflection:

$$\theta(t) = \theta_0 + b(1 - e^{-3t}) + s(1 - 0.65t) + 0.012t\sin(7t + 6u_3)$$

A strand broadens and narrows through half-width w(t), while opacity a(t) fades with distance. w_0 and a_0 set the initial scales:

$$w(t) = w_0[0.7 + 0.6\sin(\pi t)](1 - 0.7t), \quad a(t) = a_0 e^{-1.2t}(1 - t)^{1.3}$$

There are 96 angular bundles, each containing three neighboring strands. Each strand runs through 24 intervals in normalized coordinate t, from zero to one. Repeatable index hashes provide u_1 through u_4. The starting angle is $\theta_0=(k+.8(u_3-.5))2\pi/96$, bend is $b=.70(u_4-.5)$, and strand offset is $s=(j-1)(.009+.012u_2)$.

The base half-width w_0 is $R(.008+.013u_2)$, multiplied by 1.35 for the central strand and .65 for the others. The base opacity a_0 is $A(24+38u_2)$, multiplied by .8 for the central strand and .55 otherwise.

Across each strand, opacity rises from zero at the edges to a(t) at the center. Radial fading softens the end, and curvature and offsets form the hairlike bundles. The pattern stays fixed with respect to simulation time and camera orientation.

This illustrates filamentary coronal structure without reconstructing magnetic fields or predicting the appearance of a particular eclipse. The thin red chromospheric rim and small prominence markers are separate overlays.

34. Generated regions and star systems through Ctrl+P

Source for these calculations: fictionalstars_ui.cpp; fictionalstars.cpp; object_name_pool.dat; galaxygeometry.h.

Populate Region, opened with Ctrl+P, constructs a synthetic catalog from the selected region geometry, host galaxy, stellar population, spacing, and reproducible random state. Catalog numbers remain unique within the selected prefix.

Positions, stellar properties, multiple systems, orbital planes, names, and planets use independent random streams. Changing one part of generation is therefore less likely to alter unrelated parts accidentally.

To sample a spherical region without a preferred direction, cos θ and azimuth φ are uniform. The cubic-root radial law then gives uniform volume density out to radius R_{max}, before population weighting is applied:

$$\cos\theta = 2U - 1, \quad \phi = 2\pi U, \quad r = R_{max}U^{1/3}$$

Uniform directions together with $r=R_{max}\ U\text{^}(1/3)$ give uniform sampling by volume. Candidates can then be rejected according to the desired spatial density and minimum spacing. Galaxy images may supply local population weights.

Without that image information, the analytic disk density is $\exp(-3R/R_g)$ times .85 $\exp(-|z|/h)+.15\ \exp(-|z|/(3h))$; a spheroid uses $\exp(-5r/R_g)$. Clusters add further spatial scenarios. The resulting population is an illustration, not a complete Galactic survey corrected for observational selection.

For a disk-like galaxy, circular streaming speed v_c rises from the center toward scale v_0. Cylindrical radius R is compared with turnover radius R_0; the galaxy-size parameter R_g sets the nominal speed scale:

$$v_c = v_0 \frac{R}{\sqrt{R^2 + R_0^2}}, \quad v_0 = C[220\sqrt{R_g/50000}\,; 25, 320]\,\mathrm{km\,s^{-1}}$$

Random stellar motions are added around that streaming velocity. Their radial dispersion σ_R grows with age t in Gyr, while σ_z and σ_φ set vertical and azimuthal dispersions:

$$\sigma_R = 0.16 v_0 [(t + 0.1)/4.7]^{0.35}, \quad \sigma_z = 0.55\sigma_R, \quad \sigma_\phi = \sigma_R/\sqrt{2}$$

R_g is the galaxy scale in light-years, and R_0=max(100,.03R_g) sets the central turnover of the circular-speed curve. For a spheroid, the model replaces the disk dispersions with isotropic dispersion σ_R=.55v_0.

Gaussian velocity components are drawn with the Box–Muller method. Disk streaming includes an asymmetric-drift proxy; spheroids stream at .1v_c. These assigned kinematics are not evolved in a self-consistent Galactic gravitational potential.

Age and spectral-phase draws depend on the selected population and its blue-light weighting. A main-sequence lifetime proxy of $10m^{-2.5}$ Gyr limits how long high-mass stars remain in that phase. Typical main-sequence properties come from spectral tables; evolved stars receive temperature and radius distributions, with luminosity calculated from those values.

The generator does not evolve every synthetic star individually. Default rotation is clip(25√(t/4.6)(T⊙/T)²,.25,150) days. Companion mass ratios and orbital scales are drawn separately, usually near a common plane, with eccentricity below .5.

Origin of the names

Optional proper names come from the UTF-8 object_name_pool.dat file: a curated collection of mythological, historical, literary, and ancient-place names. Blank lines, comment lines, and duplicates are excluded. The pool is shuffled deterministically, and reserved names are avoided.

If the pool is exhausted, numerical suffixes allow reuse. Component and planet letters extend the host designation. These names label generated objects; they are not IAU-approved discoveries or identifiers taken from an observed star catalog.

35. Generated planets, moons, rings, and fallback properties

Source for these calculations: fictionalstars.cpp; planet.cpp; moon.cpp; orbit_spacing.h.

Planet masses are drawn log-uniformly: equal intervals in log mass have equal probability. The allowed interval extends from M_{min} to an upper limit M_{max} tied to both Jupiter's mass and the host-star mass M_*:

$$M = M_{min}(M_{max}/M_{min})^{U}, \quad M_{min} = 0.04M_{\oplus}, \quad M_{max} = \min(3M_J, 0.0003M_*)$$

For selected hosts, the regional generator requests three through ten planets. It begins with a broad mass draw and an approximate radius. In Earth units, the initial radius is m^.28 U(.9,1.1) below two Earth masses, m^.35 U(1,1.9) below twenty, and min(12,3.5(m/20)^.35) U(.8,1.2) above that.

These distributions are starting assumptions. The later classification and atmosphere calculations can revise inferred properties.

The innermost orbit must clear the star, and neighboring planets require additional separation. a_{in} sets the inner scale; h is a dimensionless spacing allowance based on the two neighboring masses relative to the star:

$$a_{in} = \max(0.02\text{AU}, 8R_*), \quad h = 8\left[\frac{M_i + M_{i-1}}{3M_*}\right]^{1/3}$$

Given the preceding orbit $a_i{-}1$, the next semimajor axis a_i allows room for both eccentric excursions and the spacing factor h. A further random multiplier widens the separation:

$$a_i = a_{i-1}\frac{1 + e_{i-1} + h}{1 - e_i - h}U(1.05, 1.30)$$

Initial eccentricities are U(0,.02), inclinations are U(0,2°) about the selected plane, and the remaining orbital angles are uniform. The first orbit lies at a_{in} U(1,2).

The outer limit is at most 500 AU. A companion can reduce it further to .05 of companion pericenter distance times a mass-ratio cube root. These spacing filters reduce clear overlaps and some unstable-looking configurations, without certifying stability by many-body integration.

Moon mass M_s is biased toward small values through U^4 and scaled to parent mass M_p. Its approximate radius R_s follows a constant-density-like cube-root mass relation with additional variation:

$$M_s = U^4 M_p/4000, \quad R_s = R_{\oplus}(M_s/M_{\oplus})^{1/3}U(0.9, 1.5)$$

Moons receive albedo U(.1,.6), color index U(.8,1.8), and eccentricity $U(0,.1)^{10}$. The first orbit is the safe inner bound multiplied by $1.1+20U^4$; successive period ratios are U(1.8,2.2).

Inclination scales as $U^4 \times .2 \times \pi/2$ and obliquity as $U^{10} \times \pi/2$, modified by an alignment factor for proximity to tidal disruption. A 3% draw makes an orbit retrograde. Clearance and outer-stability filters determine whether a candidate is accepted.

The initial ring-occurrence score is .65 for massive giants, .15 for intermediate planets, and .02 for terrestrial masses. It increases by .25 below 170 K, decreases by .60 above 300 K, and is rejected above 600 K; otherwise it is clipped to [.01,.95]. Slow or resonant rotation suppresses generation. These are appearance assumptions, not measured ring-occurrence rates.

Ring material density is 1000 kg m^{-3} below 200 K and 2500 otherwise. Inner radius is uniform from 1.1R to .6 of the Roche radius; outer radius is uniform from 1.15 times that inner radius to .98 of the Roche radius. Opacity is U(.1,.8)+.2 times normalized ring width, capped at one.

The radial texture has distinct inner and outer endpoints. Clamping at those endpoints prevents an opaque outer edge from wrapping into a transparent inner gap, including when the rings cast shadows. The reference mesh maps its normalized radial coordinate to longitudes from $-\pi$ to just below π.

When a radius is absent outside the regional generator, the general fallbacks use other relations. Solid worlds use 1.02m^.27 Earth radii; intermediate volatile-rich worlds use max(2,.56m^.67); hot giants use a sphere with an assumed density. The remaining giant fallback is 18.6 Earth radii.

That final value is a coarse visual substitute, rather than a universal giant-planet mass–radius relation. A supplied radius or a supported envelope estimate is preferable.

36. Procedural atmosphere mixtures and random reproducibility

Source for these calculations: celestial.cpp; planet.cpp; noise.cpp.

A supplied atmospheric mixture or a conserved inferred volatile inventory takes priority over an illustrative mixture. Where procedural mixtures are permitted, trace gases are drawn independently from uniform ranges and the principal gas fills the remainder. Normalization makes all fractions nonnegative and their sum one.

The table lists those starting ranges. They are assumptions about plausible-looking mixtures, rather than equilibrium chemistry or evidence for the listed gases.

Scenario	Principal draws before normalization
Gas giant	He .01–.20; CH_4 .002–.005; NH_3 .0001–.0003; C_2H_6 5–8 ppm; H_2O 0–5 ppm; remaining H_2
Ice giant	He .10–.30; CH_4 .001–.003; NH_3 .0001–.0003; C_2H_6 5–8 ppm; H_2O 0–5 ppm; H_2 .70–.85
CO_2-rich	N_2 .01–.10; SO_2 .0001–.001; Ar and H_2O 10–100 ppm; H_2S 1–10 ppm; CO 10–30 ppm; He 10–20 ppm; remaining CO_2
Cold N_2-rich	CH_4 .01–.10; H_2 .001–.005; C_2H_6 .00001–.01; remaining N_2

The optional terrestrial biosphere mixture draws CH_4 from 0–.0005, HCN from 0–.001, NH_3 from 0–10 ppm, Ar from 10 ppm–.005, CO_2 from 10 ppm–.01, and H_2O from .001–.015. CO is tied to CO_2 and ethane to methane.

A volcanic draw with probability 40% adds SO_2/H_2S; a separate 3% draw adds O_2 and proportional O_3. These probabilities are artistic settings, especially the oxygen frequency. The mixture does not follow a biological oxygenation history.

Two recurring transformations illustrate the distribution choices. The Box–Muller expression produces a standard Gaussian variate Z from independent uniform draws. X_{log} gives a log-uniform draw between positive limits l and h:

$$Z = \sqrt{-2\ln U_1}\cos(2\pi U_2), \qquad X_{log} = \exp[\ln l + U\ln(h/l)]$$

Generation uses several distributions because each gives a different kind of variation. Uniform draws weight equal intervals equally, Gaussian draws cluster around a mean, and log-uniform draws weight equal multiplicative intervals equally. Discrete choices and powers of U provide other shapes.

Integer hashes keep local variation tied to object or geographic identity. Mersenne Twister streams repeat a realization for a fixed input state and operation order, although every distribution implementation does not promise identical bits on all platforms. A reproduction record should retain source digest, catalog version, random state, epoch, weather revision, and explicit overrides.

37. Stellar surface textures and further appearance estimates

Source for these calculations: celestial.cpp: stellar-map generation; star.cpp; wdlimb.h; lavastatic.h; lavafissures.h.

Stellar activity is related to how rotation compares with convective motion. The approximate turnover time τ_{conv} depends on temperature; the Rossby number Ro is spin period divided by that time:

$$\tau_{conv} = 12[C(T; 3000, 7000)/T_{\odot}]^{-2.5} \text{ days}, \quad Ro = P_{spin}/\tau_{conv}$$

The activity amplitude A grows as Ro decreases, with bounds to limit the extrapolation. The temperature deficit ΔT_spot sets how much cooler a spot is than the surrounding photosphere:

$$A = \min(1, 0.35/\max(0.05, Ro)), \quad \Delta T_{spot} = \max[150, \min(0.541T - 1323, 0.45T)]$$

Spots and granulation can be added to ordinary stars with cool envelopes below 7000 K. Stars below .35 solar masses are treated as fully convective. Activity A gives floor(7A+U(.8,1.5)) spot groups, each with two through five members.

Low Rossby number can favor polar groups. Otherwise fully convective stars are sampled uniformly by area, while other stars favor active latitudes ±5–35°. Initial spot angular size is .35+7A degrees, modified for each member by random factors. The pattern illustrates stellar activity rather than reconstructing a magnetic field.

Granule dimensions are tied to a photospheric pressure scale height H_{photo}. The estimate uses Boltzmann's constant k_B, atomic mass unit m_u, gravity g, and an assumed particle-mass factor 1.3; the granule diameter cannot exceed half the star's radius:

$$H_{photo} = \frac{k_B T}{1.3 m_u g}, \quad d_{granule} = \min(10H_{photo}, 0.5R)$$

A correction factor of eight adjusts the visible granule count, and angular sampling limits the smallest resolved cells. Temperature contrast starts at .035T times a bounded square-root temperature factor. Larger-scale network and facular brightening depend on the activity setting.

Local temperature changes color through Planck B/V ratios. Gravity darkening uses exponent .08 for convective envelopes and .25 for radiative envelopes, together with a bounded oblateness proxy. These fields illustrate surface structure without solving radiative magnetohydrodynamics.

Limb darkening reduces intensity toward the apparent edge of a stellar disk. Here μ is the cosine of the angle between the surface normal and the observer, and a and b are fitted coefficients. Normalization by I(1) sets the disk-center intensity to one:

$$I(\mu)/I(1) = 1 - a(1 - \mu) - b(1 - \mu)^2$$

The quadratic limb-darkening coefficients depend on temperature and gravity. Published white-dwarf atmosphere colors and limb prescriptions can take priority for those objects.

Glare and coronal overlays help make features visible; they do not predict a calibrated telescope image. Likewise, glowing ground fissures and their thermal modulation remain procedural unless a measured map is explicitly supplied.

38. Periodic volatile exchange and cloud illumination

Source for these calculations: shore.cpp; cloudlighting.h.

Over an eccentric orbit, seasonal temperature changes can exchange material between gas and condensate reservoirs. The global exchange model samples 256 equal-time orbital phases and solves a first-order thermal response in normalized emitted flux.

Within each interval, the forcing is held constant and the exponential response is evaluated exactly. Over a complete cycle, the final state is a linear function plus a constant of the starting state. Requiring the end of the orbit to match its beginning therefore determines the periodic starting value directly.

The normalized thermal state f relaxes toward forcing H_j over interval Δt with response time τ. Evaluating the state halfway through the interval gives temperature through the fourth-root radiation relation, relative to mean scale T_m:

$$f_{j+1} = f_j + (H_j - f_j)(1 - e^{-\Delta t/\tau}), \quad T_j = T_m f_{j+1/2}^{1/4}$$

H_j includes a fixed contribution from internal heat and a stellar contribution that varies with orbital distance. Once T_j is known, saturation partitions the same conserved three-species inventory at that temperature.

The actual gas columns approach the equilibrium partition over finite exchange times: three days for condensation and thirty days for evaporation. These are assumed mixing times. This particular global model does not calculate cloud microphysics or return phase-change latent heat to the thermal response.

Gas column q follows a second relaxation, toward saturation target $q_{eq,j}$ over exchange time τ_{ex}. The condition $q_{end}=q_{start}$ makes the atmosphere repeat after one orbit:

$$q_{j+1} = q_j + (q_{eq,j} - q_j)(1 - e^{-\Delta t/\tau_{ex}}), \quad q_{end} = q_{start}$$

A bounded fixed-point search finds a periodic gas state for each species. It conserves the available inventory while allowing gas abundance at a given temperature to differ between warming and cooling.

Interpolation provides current gas and condensate columns, pressure, and the fractions of time categorized as frozen, condensed-liquid, or entirely accessible water vapor. Those global categories supplement rather than replace the local ocean enthalpy calculation.

Cloud illumination and precipitation phase

Cloud appearance combines a direct-light factor d with diffuse-background factor b. F_{source} uses the renderer's source-flux scale, F_* is relative stellar irradiation, and $A_{ambient}$ is ambient illumination:

$$d = C[2.5 \times 10^{-11} F_{source} / C(\sqrt{F_*}; 0.01, 1); 0, 1], \quad b = C(A_{ambient}; 0, 1)^2$$

Those factors give the day-facing cloud illumination L_{day} and the softened body-light factor L_{body}. The numerical powers and coefficients control display contrast:

$$L_{day} = \min(1, d + b), \quad L_{body} = 0.71 \min[1, 1.3 d^{0.666} + b]$$

F_{source} is expressed in the renderer's photometric units. The coefficient adapts that scale for display and is not a conversion from SI bolometric flux. Dividing by the square root of instellation helps preserve visible cloud contrast at different orbital distances.

The direct-light factor d and diffuse-background factor b are kept separate. Background starlight can therefore illuminate a cloud without being treated as direct sunlight.

Higher pressure makes diffuse sky light more important in the cloud tint. The factor f_{filter} increases logarithmically with pressure relative to Earth and is bounded between zero and one:

$$f_{filter} = C[\log_{10}(\max(1, p/p_\oplus))/2; 0, 1]$$

The filtering factor blends normalized transmitted light with normalized sky color as haze becomes stronger. It grows from terrestrial pressure to 100 terrestrial pressures, and an opaque deck forces the fully diffuse case. A sunlit cumulus lobe can reach a factor of 1.25 before clipping.

Precipitation is displayed with snow fraction $1-\text{clip}((T-273.15)/2,0,1)$, allowing a mixture just above freezing. This is a phase-appearance rule. It does not calculate precipitation chemistry or a vertical melting layer.

39. Resolved cloud lobes, wisps, and local scenery

Source for these calculations: clouds_gpu.cpp; surfacescatter.h; surfacebiomes.h; visuals.cpp; lavastatic.h.

Seen from near the surface, clouds have resolved volume as well as map-scale coverage. The opacity map supplies a local coverage measure, .29R+.53G+.18B, from which the model reconstructs a set of rounded lobes.

Blending toward an opaque deck rises between .70 and .90 coverage. Increasing angular footprint blends the detailed lobes into an unresolved cloud bank over .025–.16 in the renderer's cloud coordinates.

The local cloud-mass proxy m sets cluster scale s, while index j supplies each parent lobe's angular position θ_j. The angular increment spreads the lobes around the cluster:

$$s = 0.38 + 0.72m, \quad \theta_j = 2.39996323j + 6m$$

The variable m is a local cloud-mass proxy in display units, not a mass in kilograms. It sets the cluster size s. Each cluster contains six parent ellipsoids: one at horizontal radius zero and height −.08s, and the others at horizontal radius .43s and height (.16+.13 sin(1.7θ))s.

The central ellipsoid has horizontal and vertical axes .65s and .38s. Peripheral axes are (.43+.05 sin(2.3θ))s and (.38+.06 cos(1.9θ))s. The angular increment of 2.39996323 rad distributes the lobes around the cluster without a conspicuous regular ring.

Eight directions around each parent ellipsoid are generated from index k. The coordinate z_k distributes samples in height, and θ_k supplies azimuth, giving unit direction u_k:

$$z_k = 1 - 2(k + 0.5)/8, \quad \mathbf{u}_k = (\sqrt{1 - z_k^2}\cos\theta_k, \sqrt{1 - z_k^2}\sin\theta_k, z_k)$$

Each parent ellipsoid carries eight billows. Their centers are displaced by .55 times the parent axes along u_k, and relative radius is .52+.035 sin(2.7θ_k+j), with a small vertical adjustment. Each billow has four finer lobes, using directional offset .40 and radius .67+.025 sin(1.7θ).

Lower billows are compressed while their tops stay in place, giving staggered flat bases. The cloud floor is −s times a smooth blend from .80 to .20 over m=.35–.70. Ray–ellipsoid quadratic intersections supply silhouettes and distances; noise and a taper based on projected pixel size soften the edges.

Resolved lobes are combined with the background cloud bank through an opacity union. α_{solid} is the local lobe opacity, and α_{bank} is the unresolved contribution behind it:

$$\alpha = \alpha_{solid} + (1 - \alpha_{solid})\alpha_{bank}$$

The opacity union retains an unresolved bank behind the soft edges of resolved lobes. Small-scale geometry therefore cannot open artificial holes through an otherwise opaque cloud mass.

Wisps use a circular longitude flow, a bent crosswind latitude coordinate, and value noise at transverse scales 18 and 43. Wispiness is .001×10^(3w), where w is a clipped spherical

gradient-noise value in [0,1]. Thresholds for veils, strands, and fibers bound cirrus opacity, and cirrus fades as coverage rises from .55 to .80. These are geometric cloud shapes rather than a simulation of turbulence or droplet transport.

Scenery sizes, boundaries, and local waves

Individual scenery sizes lie between L_{min} and L_{max}. Using U^3 concentrates the sample toward the small end of that interval:

$$L = L_{min} + (L_{max} - L_{min})U^3$$

The U^3 size distribution favors small rocks and plants while retaining occasional large instances. Candidate sites come from six cube-face grids mapped onto the sphere, and their geographic identity keeps them fixed as the camera moves.

For crown exclusion, chord distance squared is compared with $(r_i+r_j+\text{gap})^2$. The lower numerical site identity wins a conflict, with a default gap of .1 m. The procedure supplies repeatable populations without modeling ecological self-organization.

Local boundaries are made irregular by displacing position r with noise before sampling a biome. The vector field n is evaluated at two different spatial scales, with the coefficients setting displacement amplitude:

$$\Delta\mathbf{r} = 32\mathbf{n}(\mathbf{r}/160) + 6\mathbf{n}(\mathbf{r}/36)$$

Before local biome lookup, Cartesian coordinates receive independent noise displacements with spatial scales 160 and 36 m and amplitudes 32 and 6 m. The displaced position is normalized back onto the sphere and converted to latitude and longitude.

This breaks up overly smooth material boundaries while avoiding seams at the poles. It changes an appearance pattern without computing erosion or the spread of vegetation.

The local ocean display uses two traveling sinusoidal components. u and v are local display coordinates and t is animation time; their weighted sum W supplies a moving crest pattern:

$$W(u,v,t) = 0.6\sin(0.6u + 0.3v - 2.4t) + 0.4\sin(-0.3u + 0.7v - 1.8t)$$

The two-wave field shifts projected ocean rows by an amplitude set in screen space. Highlights appear above a crest threshold, with higher-frequency sinusoidal sparkles added for detail. Time follows the display animation.

This is a visual wave pattern. It has no wind-driven spectrum, gravity-wave dispersion relation, or bathymetric shoaling calculation, and cannot be used to infer sea state or wind speed.

Local glowing fissures follow repeatable segmented paths. Their starting radius is 2.2+.18×view_range×U^1.8, with a perturbed heading and fourteen through nineteen segments. Main-path turns span ±48° and branch turns ±35°; branches start at segments 3, 7, and 11.

Brightness varies as .88+.12 sin(2.3t+phase). The result depicts cracks and molten material without integrating stress or magma flow.

40. Numerical coefficients for stellar and formation inference

Source for these calculations: star.cpp; stellarevolution.h; chemistry.cpp; classification_model.json.

The table below records the implemented bolometric-correction coefficients for the central stellar temperature range. Set $x=\log_{10}(T/K)$, then read each row from the constant coefficient upward: the first number multiplies x^0, the next x^1, and so on. The entries reproduce the implementation rather than introduce a new fit.

Range	Coefficients of x^0, x^1, …
$x < 3.70$	−19053.7291496456; 15514.4866764412; −4212.78819301717; 381.476328422343
$3.70 \le x < 3.90$	−37051.0203809015; 38567.2629965804; −15065.1486316025; 2617.24637119416; −170.623810323864
$x \ge 3.90$	−118115.450538963; 137145.973583929; −63623.3812100225; 14741.2923562646; −1705.87278406872; 78.8731721804990

Outside the specified temperature range, the low-temperature and above-50000 K continuations described earlier replace this polynomial. Within the main-sequence table, a continuous spectral index provides interpolation; spectral parsing and endpoint rules determine the valid interval.

The stellar evolution-grid comparison first measures a normalized distance d between an observation and a candidate model. The temperature and luminosity differences are logarithmic, and the metallicity difference is in dex:

$$d^2 = \left(\frac{\Delta \log_{10} T}{.035}\right)^2 + \left(\frac{\Delta \log_{10} L}{.20}\right)^2 + \left(\frac{\Delta [Fe/H]}{.40}\right)^2$$

Each model receives weight w based on that distance and optional mass and age penalties P_M and P_t. d_{min} is the smallest grid distance; the remaining terms express the adopted age and initial-mass population weighting:

$$\ln w = -\tfrac{1}{2}[d^2 - d_{min}^2 + P_M + P_t] + \ln(10)(\log_{10} t_{yr} - 9) - 2.35 \ln M_{initial}$$

The evolution-grid weight combines distance in the Hertzsprung–Russell diagram with optional mass and age penalties. Ordinary stellar phases also receive a constant-linear-time star-formation prior and a Salpeter initial-mass factor. The mass penalty uses squared logarithmic mass residuals with scale .30, or .04 for the compact-remnant grouping; the age penalty uses .30 dex.

Mass and age are summarized by medians. The .1 and .9 quantiles of model age bounds define the reported range, and summed weights select the phase. Cooling tracks omit the ordinary population prior. These summaries describe the adopted weighting procedure, rather than a universal Bayesian posterior for a star.

The cool-formation prior uses another log-odds score z_f. It combines planet mass m in Earth units, irradiation F, host metallicity, and a chemical offset Δ_{chem}. F_c and F_s define the onset and scale of the irradiation penalty:

$$z_f = 2\ln(m/4.37) - 0.5\ln[1 + \max(0, F - F_c)/F_s] + 3[Fe/H] + \Delta_{chem}$$

For cool-formation envelope scenarios, z_f is a log-odds score. The current settings place the mass transition at 4.37 Earth masses, with coefficient two for logarithmic mass, three for metallicity, and .5 for irradiation. F_c=1 and F_s=100 in Earth-flux units.

The chemical offset Δ_{chem}, clipped to ±1.5, adjusts the iron-only estimate using refractory enrichment and a water-rich alternative. Applying the logistic function gives a scenario score. Where calibrated flux ranges overlap, it joins the empirical classifier with balanced classes; elsewhere it remains an explicit formation assumption. The JSON settings file appears in the digest appendix because these coefficients and limits can change.

41. Worked examples and sensitivity

Source for these calculations: equations in this reference; implementation inputs stated below.

Terrestrial mass and radius

Consider a body with approximately Earth's mass and radius. Substituting those inputs into the gravity and escape-speed relations gives familiar reference values. Gravity below is expressed in m s^{-2}:

$$M = 5.9722 \times 10^{24}\ \text{kg}, \quad R = 6.371 \times 10^{6}\ \text{m} \quad \Rightarrow \quad g \simeq 9.82, \quad v_{esc} \simeq 11.19\ \text{km}\,\text{s}^{-1}$$

Now place a gas mass equal to one millionth of the planet's mass over that surface. Its hydrostatic weight gives the following pressure scale:

$$f_{atm} = 10^{-6} \quad \Rightarrow \quad p = f_{atm}\frac{GM^2}{4\pi R^4} \simeq 1.15 \times 10^{5}\ \text{Pa}$$

An atmospheric mass fraction of 10^{-6} gives a useful hydrostatic scale for an Earth-sized body. It is not the result of the full Earth calculation: release, escape, weathering, condensation, and supplied constraints change how much gas is accessible.

With unit stellar flux and A_B=.30, the equilibrium temperature is about 254.6 K. Supplying gray optical depth τ=.35 then raises the estimate to approximately 287.7 K before other constraints. The example prescribes τ to isolate its effect; it does not infer τ from a gas mixture.

Water and relief

For the same mass and radius, a surface-water mass fraction of 2.3×10^{-4} gives the following equivalent global depth. This is a volume-per-area measure before basin geometry is considered:

$$f_{W,\,surface} = 2.3 \times 10^{-4} \quad \Rightarrow \quad d_W \simeq 2693\ \text{m}$$

For an RMS Gaussian relief of 2500 m, the depth above gives $z\Phi(z)+\varphi(z)=1.077$. Solving yields z approximately .99 and coverage approximately .84. Those values follow from the assumed elevation distribution; Earth's measured basins have another geometry.

Ten times the water inventory would not give ten times the coverage. Area fraction stops at one, whereas water can continue accumulating as greater depth.

Vegetation scaling

The pressure factors can be inspected without assuming that a habitat is suitable for life. The table evaluates p_v and its crown multipliers from vacuum to 100 times terrestrial pressure:

Pressure in Earth units	p_v	Crown radius multiplier	Vertical/horizontal ratio
0	0.250	10.000	0.200
0.1	0.250	10.000	0.200
1	1.000	2.500	0.800
10	3.459	0.723	2.768
100	4.000	0.625	3.200

Every entry remains positive and bounded, including at $p=0$ and $p\oplus/e$. The table checks the geometry of the default pressure scaling. It does not imply that plants grow in vacuum: the separate habitat and life criteria still determine whether vegetation is placed.

Scientific interpretation

The formulas also show which inputs can dominate an estimate. At fixed albedo, T_{eq} varies as F^(1/4), so 10% more stellar flux raises equilibrium temperature by about 2.4%. For fixed atmospheric mass fraction, column weight varies as M^2/R^4. At fixed condensate mass, cloud optical depth varies inversely with particle radius.

Uncertain radius, inventory, aerosol size, or albedo can therefore matter more than a small improvement in another catalog measurement. Interpreting a result calls for attention to those sensitivities as well as to the displayed precision.

42. Audit changes and scientific limitations

Source for these calculations: audit edition, 13 November 2026.

The original 13 November audit accompanied two behavior corrections. Vegetation crown scaling was changed to positive, bounded environmental factors because the earlier pressure relation diverged and changed sign below $p_\oplus/e$. Low-gravity trunk scaling was also made finite. Separately, a Mercury-like 3:2 spin state was corrected to rotation period $2P_{orb}/3$, rather than $1.5P_{orb}$.

Those default-estimation corrections retain supplied nonzero morphology settings and supplied rotation periods. The present revision rewrites the explanation; it does not introduce another physical calibration.

The original audit recorded passing results for two added morphology/rotation regression tests and all four existing thermal-opacity tests. Of nineteen focused volatile, rotation, and morphology checks, eighteen passed.

The remaining cold-moon check produced approximately 244575 Pa for its Titan-like case, compared with a target of 120000–180000 Pa, using an inferred host age near 5.85 Gyr. That test does not invoke the altered default rotation or morphology calculations. The volatile solver and thermal-opacity calibration were left unchanged, and the pressure mismatch was recorded as a calibration limitation. These are the historical audit results, rather than a claim that this prose revision repeats the entire scientific audit.

The Titan-calibrated ethane term was retained as an effective net cooling adjustment. Molecular ethane itself has infrared absorption; its negative contribution in this fitted model is not a claim about negative molecular opacity.

A physical replacement would separate shortwave aerosol absorption and scattering from molecular longwave absorption and the vertical temperature profile, then validate their combined energy budget. Retaining the old calibration avoids introducing an unvalidated change to that coupled approximation.

Several approximations recur across the models. Irradiation follows one principal stellar source. Bolometric corrections and Bond albedo can be extrapolated beyond sparse calibrations. Infrared warming uses gray coefficients, and volatile escape is represented by effective rates. Basin relief and interior storage are assumed, while saturation uses constant latent heats and water lacks a general high-pressure phase diagram.

Weather patterns are procedural, biosphere probabilities are speculative, and some rock and plant forms depend on artistic environmental rules. These qualifications limit what the generated properties can establish. They should accompany any scientific use of the results, rather than be replaced by a nominal uncertainty interval.

This reference explains the model families used to infer properties and construct appearance. Decorative constants, display-sampling choices, and synthetic names do not all have a physical derivation.

The source locations and digests that follow identify the implementation present when the document is built. They allow a later reader to distinguish that repository state from the original audit and from subsequent revisions.

43. Primary sources and calibration context

Source for these calculations: published articles and reference data, checked 13 November 2026.

The references identify the observational data and published methods behind specific parts of the model. Each entry explains its role in the application; the application's extra assumptions are described in the corresponding sections above.

R1. Zahnle & Catling (2017), The cosmic shoreline. This paper motivates the atmospheric-retention trend and the use of XUV exposure as a historical factor.

https://arxiv.org/abs/1702.03386

R2. Lopez & Fortney (2014), Understanding the Mass–Radius Relation for Sub-Neptunes. Their envelope-thickness fit supplies the radius relation; its published domain is stated in the envelope section.

https://arxiv.org/abs/1311.0329

R3. Eisenman & Wettlaufer (2009), Nonlinear threshold behavior during the loss of Arctic sea ice. This work motivates the enthalpy and conductive-ice treatment. The parameters adopted for other worlds remain application scenarios.

https://arxiv.org/abs/0812.4777

R4. NIST Chemistry WebBook, SRD 69, ethane infrared spectrum. These data establish that ethane absorbs infrared radiation. They do not prescribe the fitted net climate response used in the application.

https://webbook.nist.gov/cgi/cbook.cgi?ID=C74840&Index=0&Type=IR-SPEC

R5. Torres (2010), On the Use of Empirical Bolometric Corrections for Stars. This supplies the corrected Flower coefficients used for the stellar bolometric correction.

https://arxiv.org/abs/1008.3913

R6. Leconte et al. (2010), Is tidal heating sufficient to explain bloated exoplanets? This is the reference for the full-eccentricity equilibrium-tide functions discussed in the heating section.

https://doi.org/10.1051/0004-6361/201014337

R7. Santos et al. (2017), Constraining planet structure from stellar chemistry. This work motivates estimating condensate proportions through stoichiometric abundance bookkeeping.

https://arxiv.org/abs/1711.00777

R8. NIST Chemistry WebBook, nitrogen phase-change data, together with the Fray & Schmitt solid–vapor fits reproduced by Marboeuf et al. (2014). These provide the phase-change context for the nitrogen saturation relations.

https://webbook.nist.gov/cgi/cbook.cgi?ID=C7727379&Mask=4

R9. Kopparapu et al. (2014), Habitable Zones Around Main-Sequence Stars: Dependence on Planetary Mass. The application uses the polynomial limits and coefficients listed in Table 1.

https://arxiv.org/abs/1404.5292

R10. Cahoy, Marley & Fortney (2010), Exoplanet albedo spectra and colors as a function of planet phase, separation, and metallicity. This work provides the context for the giant-planet reflection library.

https://doi.org/10.1088/0004-637X/724/1/189

The cited works support the particular methods or data named beside them. They do not independently validate every default, extrapolation, appearance probability, or rendering choice in Alienorum. The equations and application-specific qualifications in this reference describe how those ingredients are actually used.

44. Source traceability and reproduction

Source for these calculations: repository files at document build time.

Each source file below is identified by the first sixteen hexadecimal characters of its SHA-256 digest. These locations connect the scientific explanation to reproducible implementation details; they also identify the repository state at document build time, which can include changes made after the original audit.

To rebuild the PDF, run docs/build_calculation_reference.py from the repository root with Python 3, ReportLab, Matplotlib, and NumPy available. The document embeds DejaVu fonts and equations rendered at 330 dpi, so the reader can view the mathematical notation directly.

Source locator	SHA-256 prefix
src/classes/shore.cpp	bb430861d11558ce
src/classes/shore.h	7d60707afb2b5774
src/classes/planet.cpp	44554da2dd96de2c
src/classes/celestial.cpp	f0363aaf95e29891
src/classes/star.cpp	bf63e592517c0a21
src/classes/misc.cpp	152315c059eb1ebf
src/classes/color.cpp	ed4d822e0240d35a
src/classes/chemistry.cpp	d3d562122e729e0d
src/classes/patch.cpp	f48c22c05b1eece4
src/classes/climate.h	dd010f626f3a3f31
src/classes/thermalclimate.h	23caede86b6e3997

Source locator	SHA-256 prefix
src/classes/iceclimate.h	b1418fdda76f06d7
src/classes/salinity.h	bf8f77f28d10af5e
src/classes/surfaceclimate.h	5d9f2eab6b1288dd
src/classes/vegetation.h	5fe99eaa125f9e5b
src/classes/plantform.h	4db1840088eb8533
src/classes/plantmesh.h	63ac3e57cf715ef5
src/classes/planttexture.h	f4b9fe66f75814e7
src/classes/rockmeshcache.h	d284069523a45a06
src/classes/cloudclimate.h	efc5f2d9fdcff0c0
src/classes/giantclouds.h	a49984f107c0463a
src/classes/giantreflectance.cpp	b7cc55b65a386eb7
src/classes/pressurehaze.h	d578ef5b2f6fbb35
src/classes/surfacerefraction.h	62784b3c1a1b18e4
src/classes/tidalheat.h	4ec39223bf8411b7
src/classes/habitablezone.h	158a65287be4ce7b
src/classes/noise.cpp	4192ce7f1c23e2e9
src/classes/stellarevolution.h	2df2954763ba8a09
src/classes/cloudlighting.h	cd94a52aaa9d04c5
src/classes/surfacescatter.h	9f03ea268a2a4c7b
src/classes/surfacenavigation.cpp	e253f46700df0952
src/classes/surfacenavigation.h	8f6c330de6aecbc5
src/classes/surfacebiomes.h	7f04e553eea35922
src/classes/lavastatic.h	d61deae0e2629bf2
src/classes/clouds_gpu.cpp	776a58f308a00420
src/classes/fictionalstars.cpp	209930f91cdcd9b2
src/fictionalstars_ui.cpp	1f30bf3404cb5614
src/visuals.cpp	12ea1d3959d1110a
src/windowidle.cpp	5128c130b52049f7

Source locator	SHA-256 prefix
src/windowidle.h	d3d60f3b875209bd
src/sphere_impostor.cpp	e3a532718c9f850f
src/alienorum.cpp	30c01857fb2cbace
object_name_pool.dat	53dab0edda46bd39
catalogs/NASA/classification_model.json	031a9f167677132d
catalogs/giant_reflectance.json	d4ba2290f722102f